TU Clausthal

MASTER THESIS

An Intelligent Analysis Approach of Real Test Drive Recordings Based on Self-Supervised Learning and HIL Real-Time Simulation Dataset

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List of Abbreviations

HIL Hardware-in-the-Loop

A2D2 Audi Autonomous Driving Dataset

SimCLR Simple Contrastive Learning of Representations

NT-Xent Normalized Temperature-scaled Cross-Entropy

ROC Receiver Operating Characteristic

AUC Area Under the Curve

SSL Self-Supervised Learning

ABS Anti-lock Braking System

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Abstract

The evolution of Automotive Software Systems (ASSs) has led to growing complexity in sensor networks, increasing the risk of undetected faults that can compromise vehicle safety. Traditional fault detection methods depend on supervised learning with large labeled datasets, which are expensive to obtain and inherently incomplete—they cannot cover every possible failure mode.

This thesis proposes a self-supervised contrastive learning framework for automotive sensor fault detection that requires no labeled fault data. A 1D Convolutional Neural Network encoder is trained using the SimCLR framework on healthy driving data from the Audi Autonomous Driving Dataset (A2D2), learning to map time-series sensor windows into a 256-dimensional embedding space. Anomalies are detected by measuring the cosine similarity between test embeddings and a healthy reference centroid computed from a small calibration set of Hardware-in-the-Loop (HIL) data.

The framework is evaluated on six fault scenarios—gain, noise, and stuck-at faults injected into accelerator pedal and vehicle speed sensors—across six detection threshold percentiles (15th through 40th). Comprehensive metrics including precision, recall, F1-score, accuracy, confusion matrices, and ROC curves are reported for all experiments, alongside computing cost measurements for training and inference.

Results demonstrate that self-supervised representations learned from real-world driving data transfer effectively to HIL simulation data for fault detection, without any labeled fault examples. The multi-threshold analysis quantifies the precision–recall trade-off, providing practical guidance for threshold selection in safety-critical applications.

Keywords: Self-Supervised Learning, Contrastive Learning, SimCLR, Automotive Sensor Fault Detection, Hardware-in-the-Loop, Anomaly Detection, 1D-CNN, Transfer Learning

Zusammenfassung

Die zunehmende Komplexität moderner Fahrzeugsoftwaresysteme erhöht das Risiko unerkannter Sensorfehler, die die Fahrzeugsicherheit gefährden können. Herkömmliche Fehlererkennungsmethoden basieren auf überwachtem Lernen mit großen gelabelten Datensätzen, deren Beschaffung kostenintensiv und inhärent unvollständig ist.

Diese Arbeit schlägt ein selbstüberwachtes kontrastives Lernverfahren zur Fehlererkennung bei Fahrzeugsensoren vor, das keine gelabelten Fehlerdaten benötigt. Ein 1D-CNN-Encoder wird mittels des SimCLR-Frameworks ausschließlich auf gesunden Fahrdaten des Audi Autonomous Driving Dataset (A2D2) trainiert. Anomalien werden durch Kosinusähnlichkeit der Einbettungen zu einem gesunden Referenz-Zentroid erkannt, der aus einer kurzen Kalibrierungsaufnahme von Hardware-in-the-Loop-Daten berechnet wird.

Das Framework wird an sechs Fehlerszenarien evaluiert—Verstärkungsfehler, Rauschfehler und Stuck-at-Fehler in Gaspedalstellung und Fahrzeuggeschwindigkeit—über sechs Schwellenwert-Perzentile (15. bis 40.). Präzision, Recall, F1-Score, Genauigkeit, Konfusionsmatrizen und ROC-Kurven werden für alle Experimente berichtet, zusammen mit den Berechnungskosten für Training und Inferenz.

1 Introduction

1.1 Context and Motivation

Modern vehicles rely on a growing network of Electronic Control Units (ECUs) that continuously process data from dozens of sensors to govern safety-critical functions such as engine management, braking, and steering (Abboush, Knieke, and Rausch, 2022). A single mid-range passenger car today can contain more than 70 ECUs communicating over Controller Area Network (CAN) buses, each dependent on accurate sensor readings to maintain safe operation. When a sensor produces faulty measurements—whether due to electrical interference, mechanical degradation, or component failure—the downstream control logic may issue incorrect commands, potentially leading to hazardous situations for occupants and other road users.

The ISO 26262 standard for functional safety of road vehicles mandates rigorous testing and validation of these systems, including Hardware-in-the-Loop (HIL) simulation and real-world test drives (*ISO 26262: Road Vehicles – Functional Safety* 2018). During such validation campaigns, vast quantities of time-series sensor data are recorded, often exceeding several gigabytes per test session. Manual inspection of this data by domain experts is not only time-consuming but also prone to human oversight, particularly for subtle or intermittent faults that may not produce immediately obvious signal distortions.

Artificial intelligence offers a path toward automated fault detection. Supervised deep learning methods, including Convolutional Neural Networks (CNNs) combined with Gated Recurrent Units (GRUs), have demonstrated strong performance in classifying fault types and locations within automotive systems (Ghannoum, 2025; Abboush, Knieke, and Rausch, 2022). However, these approaches depend on large, labeled datasets that include examples of every fault type the model is expected to recognize. Collecting such labeled fault data requires deliberate fault injection campaigns and manual annotation—a

process that is expensive, time-intensive, and inherently incomplete, since it cannot cover every conceivable fault mode that may arise in practice.

Self-supervised learning (SSL) offers a fundamentally different paradigm. Rather than learning from labeled examples, SSL methods extract supervisory signals from the structure of the data itself (Jaiswal et al., 2021). In the context of fault detection, this means a model can learn what *normal* sensor behavior looks like by training exclusively on healthy data, which is abundantly available from routine test drives. Any subsequent deviation from the learned normal patterns can then be flagged as a potential anomaly, without ever having seen a single labeled fault example during training.

Among SSL approaches, contrastive learning has emerged as particularly effective for representation learning (T. Chen et al., 2020). The core principle is to train an encoder network such that similar inputs (augmented views of the same data sample) produce similar representations, while dissimilar inputs produce distant representations. SimCLR (ibid.), one of the most widely adopted contrastive frameworks, has demonstrated state-of-the-art results across diverse domains, yet its application to automotive sensor fault detection remains largely unexplored.

This thesis investigates whether contrastive self-supervised learning, trained solely on healthy driving data from the Audi Autonomous Driving Dataset (A2D2) (Audi AG, 2020), can effectively detect sensor faults injected through a HIL simulation environment. The work addresses a practical scenario: a model trained on real-world driving data from one vehicle is deployed to detect faults in a different data acquisition system (HIL test bench), testing the cross-domain generalization capability of the learned representations.

1.2 Research Questions

This thesis addresses the following research questions:

- **RQ1:** Can a contrastive self-supervised learning model, trained exclusively on healthy sensor data from real-world driving recordings, reliably detect sensor faults in HIL simulation data without any labeled fault examples?

- **RQ2:** How does the selection of the anomaly detection threshold affect the trade-off between detection sensitivity (recall) and false alarm rate (precision), and what threshold range is most suitable for safety-critical automotive applications?
- **RQ3:** Which types of sensor faults (gain, noise, stuck-at) are most and least detectable by the proposed approach, and what physical factors explain the observed differences?

1.3 Related Work

This section reviews prior research at the intersection of self-supervised learning, contrastive representation learning, anomaly detection, and automotive fault diagnosis. The objective is to situate the proposed method within the existing literature and to identify the specific gaps that this thesis addresses.

Xu et al. (2024) survey self-supervised learning methods for machinery fault diagnosis and conclude that contrastive learning is the most effective paradigm when labelled data are scarce. Their review, covering over 100 publications, reveals that the vast majority of SSL-based fault diagnosis studies focus on benchmark bearing datasets; automotive sensor systems remain almost entirely unexplored.

Wang et al. (2023) combine contrastive learning with wavelet-based time-frequency analysis for bearing fault diagnosis. The method achieves 95.3% accuracy on the CWRU dataset after fine-tuning with only 10% of the available labels. Although label efficiency is improved, the approach still requires a small labelled set for fine-tuning and has not been validated on automotive sensor signals. Similarly, Ding et al. (2022) employ contrastive predictive coding for incipient bearing fault detection, capturing temporal dependencies that enable early fault identification. Again, labelled examples are required for classifier adaptation, and the method is designed for vibration data rather than for accelerator or speed sensor signals.

J. Li, Xie, and Zong (2023) propose a semi-supervised contrastive learning framework that reduces the number of labelled examples needed for bearing fault classification to as few as ten per fault class. While this represents a substantial reduction, the dependency on any labelled fault data limits applicability in scenarios where certain fault types have never been observed.

Zhang et al. (2023) introduce a masked autoencoder that reconstructs randomly masked portions of vibration signals; the learned representations transfer well to downstream classification tasks. The generative pretext task, however, is inherently less discriminative for anomaly detection than contrastive objectives, and the evaluation remains confined to rotating machinery.

Among contrastive learning frameworks, SimCLR (T. Chen et al., 2020) established that strong visual representations can be learned by maximising agreement between augmented views of the same sample while minimising agreement with views of other samples. Its core components—data augmentation, a base encoder, a projection head, and the NT-Xent loss—are architecture-agnostic and have inspired numerous time-series adaptations. MoCo (He et al., 2020) addresses SimCLR's batch-size sensitivity by maintaining a momentum-updated dictionary of negative examples, while BYOL (Grill et al., 2020) eliminates the need for negative pairs entirely. For temporal data, TS2Vec (Yue et al., 2022) performs contrastive learning at multiple temporal scales, and TS-TCC (Eldele et al., 2021) introduces paired weak-strong augmentations. Despite these advances, no prior work has systematically adapted SimCLR to automotive sensor fault detection, nor has any study evaluated its ability to transfer representations from real driving data to hardware-in-the-loop simulated faults.

Ruff et al. (2021) provide a theoretical foundation for deep anomaly detection, demonstrating that self-supervised pretraining followed by simple distance-based scoring in the learned feature space consistently outperforms end-to-end trained detectors. Pang et al. (2021) identify the core challenges of learning effective anomaly representations without labelled anomalies and of detecting complex, multi-feature faults. Hojjati and Armanfard (2023) confirm that contrastive embeddings naturally separate in-distribution from out-of-distribution samples, supporting the two-stage pipeline adopted in this thesis: contrastive pretraining on healthy data only, followed by cosine similarity thresholding.

Within the automotive domain, Zhao et al. (2024) review deep learning approaches for sensor fault detection and explicitly identify label scarcity as the primary barrier to adoption; they recommend self-supervised and transfer learning as key future directions. Abboush, Knieke, and Rausch (2022) and Ghannoum (2025) propose supervised CNN-GRU frameworks that achieve high classification accuracy on HIL-simulated faults. Both methods, however, require labelled examples of every fault type and location. Neither operates in an unsupervised setting. J. Chen et al. (2023) survey domain adaptation methods for bearing fault diagnosis, a scenario analogous to the A2D2-to-HIL

transfer addressed in this thesis. Tang and Liu (2024) review SSL applications in automotive perception, confirming the viability of contrastive methods in the automotive domain but not for fault detection.

A synthesis of the reviewed literature reveals six recurring research gaps: (i) SSL-based fault diagnosis remains concentrated on rotating machinery; automotive sensor systems are neglected. (ii) Existing SSL methods reduce but do not eliminate the need for labelled fault data (fine-tuning, semi-supervised classification). (iii) Cross-domain transfer between real and simulated acquisition systems has not been systematically evaluated for SSL-based automotive fault detection. (iv) Detection performance is typically reported at a single operating point, ignoring the precision–recall trade-off controlled by threshold selection. (v) Many studies report only accuracy, omitting precision, recall, F1-score, and ROC analysis essential for safety-critical applications. (vi) Computational costs of training and inference are rarely reported, obscuring practical feasibility.

This thesis directly addresses all six gaps. It applies contrastive SSL to automotive accelerator and speed sensors, requires *zero* labelled fault data, validates A2D2-to-HIL transfer, evaluates six threshold settings, reports comprehensive metrics for every experiment, and measures computational costs for all pipeline stages. In doing so, to the best of our knowledge, it provides the first demonstration of fully unsupervised, cross-domain automotive sensor fault detection using self-supervised contrastive learning.

1.4 Solution

The proposed framework operates in three distinct phases:

1. **Self-supervised pretraining.** A one-dimensional CNN encoder is trained on the SimCLR contrastive learning objective using healthy driving data from the A2D2 dataset. The encoder learns to map time-series sensor windows into a compact embedding space where semantically similar windows are placed close together. Three domain-specific augmentation strategies—Gaussian jittering, amplitude scaling, and temporal masking—generate the positive pairs required for contrastive training.

2. **Anomaly detection calibration.** A small quantity of healthy HIL data is passed through the frozen encoder to establish a reference distribution in the embedding space. The centroid of the healthy embeddings and

the distribution of cosine similarities to that centroid are computed, and detection thresholds are set at multiple percentiles of this distribution.

3. **Fault detection and evaluation.** HIL data containing injected faults (gain, noise, and stuck-at types across accelerator pedal and vehicle speed sensors) is evaluated against the calibrated thresholds. Performance is assessed using precision, recall, F1-score, accuracy, confusion matrices, and Receiver Operating Characteristic (ROC) curves across multiple threshold settings.

1.5 Objective

The primary objective of this thesis is to develop, implement, and evaluate a self-supervised contrastive learning framework for automotive sensor fault detection that satisfies the following requirements:

1. Requires no labeled fault data during any phase of training or calibration.
2. Achieves reliable detection of gain, noise, and stuck-at sensor faults in HIL simulation data.
3. Provides a systematic multi-threshold analysis (15th through 40th percentile) to quantify the sensitivity–specificity trade-off.
4. Demonstrates cross-domain transfer from real-world A2D2 data to HIL data.
5. Reports all evaluation metrics mandated for safety-critical systems: precision, recall, F1-score, accuracy, confusion matrices, and ROC curves with AUC values.
6. Documents computational costs for training and inference to assess deployment feasibility.

1.6 Structure

The remainder of this thesis is organized as follows:

- **Chapter 2: Background.** Introduces the theoretical foundations, including model-based development and testing phases in the automotive V-model, the real test drive validation process, fault injection approaches, contrastive self-supervised learning with detailed mathematical formulations, and anomaly detection methods.
- **Chapter 3: Methodology.** Presents the theoretical foundations and methodological design of the proposed self-supervised fault detection framework, including problem formulation, contrastive learning with SimCLR, domain-appropriate data augmentations, 1D-CNN encoder architecture, healthy reference modeling via centroid estimation, similarity-based anomaly detection, evaluation metrics, and the cross-domain transfer setting.
- **Chapter 4: Implementation and Experimental Setup.** Describes the concrete realization of the proposed methodology, including A2D2 and HIL dataset processing, normalization and windowing pipelines, the 1D-CNN encoder and projection head implementation, hyperparameter configuration, training procedure with SimCLR, calibration of the healthy centroid and percentile thresholds, and the evaluation protocol for both per-fault and binary classification. Computational environment, reproducibility measures, and intermediate validation results are also documented.
- **Chapter 5: Results and Discussion.** Reports the full evaluation results, including per-fault detection performance across six thresholds, binary classification results (healthy vs. faulty), ROC analysis, per-fault-type analysis, and computing cost measurements. A detailed discussion interprets the findings in the context of practical automotive deployment.
- **Chapter 6: Conclusion and Future Work.** Summarizes the contributions, answers the research questions, and outlines directions for future research.

2 Background

This chapter introduces the theoretical foundations required to understand the proposed fault detection framework. It covers the automotive development and validation process, fault injection techniques used in HIL testing, the fundamentals of deep learning for time-series data, and—most critically—a detailed treatment of contrastive self-supervised learning with full mathematical formulations.

2.1 Model-Based Development and Testing Phases

Model-Based Development (MBD) has become the standard methodology in the automotive industry for designing and validating embedded software systems. MBD uses mathematical and simulation models to facilitate early design validation and verification, following the V-model development lifecycle (Abboush, Knieke, and Rausch, 2022). The left side of the V-model represents the design and development stages, while the right side represents the corresponding verification and validation stages, progressing from abstract models to physical hardware.

2.1.1 Model-in-the-Loop (MIL)

Model-in-the-Loop testing validates the system design within a purely virtual environment. Both the controller and the plant are represented as mathematical models, typically developed in tools such as MATLAB/Simulink. MIL enables early verification of algorithmic correctness, control strategy logic, and functional behavior without requiring any hardware components. Because MIL testing is entirely software-based, it offers the fastest iteration cycles and the lowest cost among all V-model testing phases.

2.1.2 Software-in-the-Loop (SIL)

In SIL testing, the controller model is replaced by the actual generated source code, which runs on the host development computer against the simulated plant model. SIL testing verifies that the auto-generated code behaves identically to the original controller model, identifying discrepancies that may arise from code generation, numerical precision, or implementation-specific behavior.

2.1.3 Processor-in-the-Loop (PIL)

PIL testing executes the controller software on the target microprocessor or ECU hardware, while the plant model still runs in simulation. This phase captures processor-specific effects such as fixed-point arithmetic, limited memory, interrupt handling, and execution timing that are invisible during SIL testing.

2.1.4 Hardware-in-the-Loop (HIL)

HIL testing represents a critical validation stage where the complete ECU hardware, including its physical interfaces, runs against a real-time simulation of the vehicle plant model (dSPACE GmbH, 2023). The HIL simulator generates realistic electrical signals that mimic the outputs of real sensors (e.g., accelerometer readings, wheel speed pulses, pedal position voltages) and receives the ECU's actuator commands in return. This closed-loop configuration allows validation of the entire hardware-software stack under conditions that closely approximate real vehicle operation, but in a safe, repeatable laboratory environment.

HIL testing is particularly valuable for fault injection studies because faults can be introduced systematically at precisely defined times, locations, and magnitudes—something that would be impractical or dangerous to do in a real vehicle. The dSPACE HIL platform, used in the research group at TU Clausthal, provides Configuration Desk for system setup and Control Desk for real-time monitoring and data logging during test execution (Abboush, Knieke, and Rausch, 2022).

2.1.5 Vehicle-in-the-Loop (VIL)

VIL testing places a physical vehicle—or a high-fidelity driving simulator—within a virtual environment that generates realistic driving scenarios. The vehicle's control systems respond to simulated road conditions, traffic, and weather, while the virtual environment adapts in real time to the vehicle's actions. VIL bridges the gap between laboratory HIL testing and real-world test drives.

2.2 The Real Test Drive Validation Process

The real test drive constitutes the final validation stage for automotive systems before production release. During these drives, the vehicle operates under actual road conditions while onboard data acquisition systems continuously record sensor measurements at high sampling rates. This section describes the process as it relates to the data used in this thesis.

2.2.1 Test Drive Data Acquisition

During a real test drive, data acquisition systems record signals from all relevant sensors and ECUs over the vehicle's CAN bus. Typical recorded signals include:

- Accelerator pedal position (%), reflecting the driver's throttle input
- Vehicle speed (km/h), measured by wheel speed sensors
- Engine speed (RPM), from the crankshaft position sensor
- Steering wheel angle (degrees), from the steering column sensor
- Brake pedal position (%), indicating braking intensity
- Lateral and longitudinal acceleration (m/s^2), from inertial measurement units

Sampling rates vary by sensor type. In the Audi A2D2 dataset used in this thesis, the accelerator pedal signal is recorded at approximately 100 Hz, while the vehicle speed signal is recorded at approximately 50 Hz. These different rates reflect the physical characteristics of each measurement: the accelerator pedal position can change rapidly with the driver's foot movement, while vehicle speed changes more gradually due to the vehicle's inertia.

2.2.2 Data Volume and Analysis Challenges

A single test drive of 30 minutes at 100 Hz sampling generates over 180,000 data points per sensor channel. With multiple sensors recorded simultaneously, the total data volume per test session can reach several gigabytes. Manual expert analysis of this volume is impractical, motivating the development of automated analysis methods.

The key challenge is that faults in real-world data are rare, unlabeled, and diverse. A sensor may produce subtly incorrect readings for only a few seconds within hours of normal operation. Traditional threshold-based monitoring catches only the most severe deviations, while intermittent or slowly drifting faults may go undetected. This motivates the self-supervised approach developed in this thesis, which learns a comprehensive model of normal behavior and can identify even subtle deviations.

2.2.3 The A2D2 Dataset

The Audi Autonomous Driving Dataset (A2D2) (Audi AG, 2020) is a large-scale public dataset recorded by Audi AG for research in autonomous driving. It contains sensor data from multiple test drives conducted in three German cities: Gaimersheim, Ingolstadt, and Munich. The dataset includes camera images, LiDAR point clouds, and CAN bus signals recorded during real-world driving.

For the purpose of this thesis, only the CAN bus signals are used, specifically the accelerator pedal position and vehicle speed. These signals provide a representative pair of automotive sensors with a clear physical relationship: the accelerator pedal is a driver input, and the vehicle speed is the corresponding system output. The data is stored in JSON format with microsecond-resolution timestamps, enabling precise sampling rate analysis and signal alignment.

2.3 Fault Injection Approaches

Fault injection is a validation technique used to assess how a system behaves in the presence of faults (Abboush, Knieke, and Rausch, 2022). ISO 26262 recommends fault injection at various stages of the V-model to verify that safety mechanisms correctly detect and mitigate faults. A fault injection scenario is characterized by three attributes: *when* the fault occurs (fault time), *where* it occurs (fault location), and *what type* of fault is introduced (fault type).

2.3.1 Fault Types

Table 2.1 summarizes the three fault types used in this thesis, along with their mathematical representations and typical physical causes.

Table 2.1: Mathematical representations of signal fault types used in this thesis.

Fault Type	**Mathematical Representation**	**Physical Cause**
Healthy signal	$y(t) = x(t)$	Normal operation
Gain fault	$y(t) = \alpha \cdot x(t), \quad \alpha \neq 1$	Calibration error, amplifier drift
Noise fault	$y(t) = x(t) + \eta(t), \quad \eta \sim \mathcal{N}(0, \sigma^2)$	EMI, sensor degradation
Stuck-at fault	$y(t) = c \quad \forall t \geq t_f$	Broken wiring, component failure

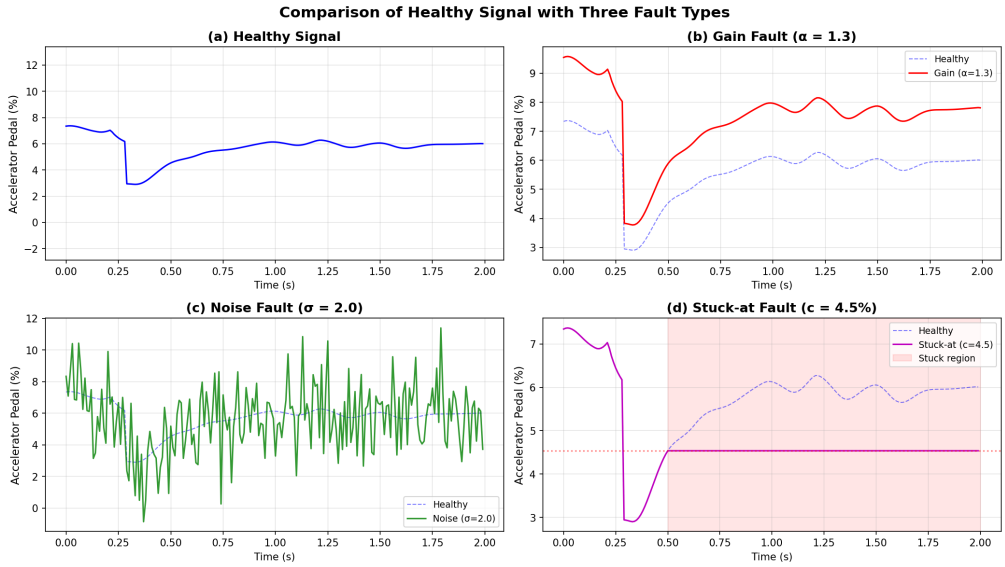

Figure 2.1: Comparison of a healthy sensor signal with three fault types: gain, noise, and stuck-at.

Gain fault. The sensor output is scaled by an incorrect factor α. When $\alpha > 1$, the signal is amplified; when $\alpha < 1$, it is attenuated. This fault typically arises from calibration errors in the signal conditioning circuit or from amplifier drift due to temperature changes. The distortion is proportional to the signal magnitude, meaning it is most visible during dynamic driving conditions and nearly invisible when the true value is close to zero.

Noise fault. Additive Gaussian noise $\eta(t)$ corrupts the sensor output. The noise standard deviation σ determines the fault severity. Noise faults are caused by electromagnetic interference (EMI) from nearby components, degraded shielding, or loose electrical connections. Unlike gain faults, noise faults are visible regardless of the signal magnitude, appearing as high-frequency fluctuations superimposed on the true signal.

Stuck-at fault. The sensor output freezes at a constant value c from the fault onset time t_f onward. This represents the most severe fault type, corresponding to a complete sensor failure such as a broken wire, shorted circuit, or failed sensing element. The frozen value may be zero, the last valid reading, or an arbitrary voltage determined by the failure mode.

2.3.2 Fault Locations

In this thesis, faults are injected into two sensor signals:

1. **Accelerator pedal position (%):** Measures the driver's throttle demand. Range: 0% (released) to 100% (fully pressed). A fault in this sensor can cause unintended acceleration or loss of throttle response.

2. **Vehicle speed (km/h):** Measures how fast the vehicle is traveling. A fault in this sensor can affect cruise control, ABS, and stability control systems.

The combination of three fault types and two fault locations yields six distinct fault scenarios, each evaluated independently in Chapter 5.

2.4 Deep Learning for Time-Series Data

This section introduces the deep learning architectures relevant to the encoder network used in this thesis.

2.4.1 Convolutional Neural Networks (CNNs)

A CNN applies learnable convolutional filters to extract local patterns from input data (LeCun, Bengio, and Hinton, 2015). For one-dimensional time-series data, a 1D convolutional layer computes:

$$z_j^{(l)}[n] = \sigma\left(\sum_{i=1}^{C_{l-1}} \sum_{m=0}^{K-1} w_{ij}^{(l)}[m] \cdot a_i^{(l-1)}[n \cdot s + m] + b_j^{(l)}\right) \tag{2.1}$$

where $a_i^{(l-1)}[n]$ is the i-th input channel at position n, $w_{ij}^{(l)}[m]$ are the filter weights of length K, s is the stride, $b_j^{(l)}$ is the bias term, $\sigma(\cdot)$ is a nonlinear activation function, and $z_j^{(l)}[n]$ is the j-th output feature map at position n. The number of input channels is C_{l-1} and the number of output channels (filters) is C_l.

2.4.2 Batch Normalization

Batch normalization (Ioffe and Szegedy, 2015) normalizes the pre-activation values within each mini-batch to stabilize and accelerate training:

$$\hat{z}_j = \frac{z_j - \mu_{\mathcal{B}}}{\sqrt{\sigma_{\mathcal{B}}^2 + \epsilon}} \cdot \gamma + \beta \tag{2.2}$$

where $\mu_{\mathcal{B}}$ and $\sigma_{\mathcal{B}}^2$ are the mean and variance computed over the current mini-batch, γ and β are learnable scale and shift parameters, and ϵ is a small constant for numerical stability.

2.4.3 Pooling Operations

Max pooling selects the maximum value within each pooling window of size p:

$$a_j^{\text{pool}}[n] = \max_{m=0}^{p-1} z_j[n \cdot p + m] \tag{2.3}$$

Global average pooling computes the mean across the entire temporal dimension:

$$\bar{a}_j = \frac{1}{T} \sum_{n=1}^{T} z_j[n] \tag{2.4}$$

producing a single scalar per feature map. This operation converts variable-length feature maps into a fixed-size representation, which is essential for mapping time-series windows of arbitrary length to fixed-dimensional embeddings.

2.5 Self-Supervised Learning

Self-supervised learning (SSL) is a paradigm in which a model learns useful representations from unlabeled data by solving pretext tasks that are derived from the data's own structure (Jaiswal et al., 2021). The supervisory signal comes not from human annotations but from the inherent properties of the data—temporal order, spatial coherence, or augmentation invariance.

SSL methods can be categorized into three families:

1. **Generative methods** learn to reconstruct the input data (e.g., autoencoders, variational autoencoders). The reconstruction objective encourages the model to capture the most informative features of the data.

2. **Predictive methods** learn to predict a missing or future part of the input from the available context (e.g., masked language models, Contrastive Predictive Coding (Oord, Y. Li, and Vinyals, 2018)).

3. **Contrastive methods** learn to distinguish between similar and dissimilar data pairs in a learned embedding space (T. Chen et al., 2020; He et al., 2020). This thesis employs contrastive learning.

The advantage of SSL for fault detection is that it requires only healthy data for training. Since healthy sensor recordings are abundantly available from routine test drives and HIL experiments, SSL eliminates the need for expensive labeled fault datasets.

2.6 Contrastive Learning

Contrastive learning trains an encoder to produce representations in which similar samples are close together and dissimilar samples are far apart. This section provides the mathematical formulation in detail, as the contrastive objective is the central training mechanism of the proposed framework.

2.6.1 Problem Formulation

Given an unlabeled dataset $\mathcal{D} = \{x_1, x_2, \ldots, x_N\}$ of N sensor data windows, the goal is to learn an encoder function $f_\theta : \mathcal{X} \to \mathbb{R}^d$ parameterized by θ that maps each input $x_i \in \mathcal{X}$ to a d-dimensional representation $h_i = f_\theta(x_i)$ such that the representation captures the essential structure of the data.

To achieve this without labels, contrastive learning constructs *positive pairs* (two views of the same sample that should have similar representations) and *negative pairs* (views of different samples that should have dissimilar representations).

2.6.2 Data Augmentation

A stochastic augmentation module $\mathcal{T}$ transforms each sample x_i into two correlated views:

$$\tilde{x}_i = t(x_i), \quad t \sim \mathcal{T} \tag{2.5}$$

$$\tilde{x}_j = t'(x_i), \quad t' \sim \mathcal{T} \tag{2.6}$$

where t and t' are independently sampled augmentation functions from the family $\mathcal{T}$. The pair $(\tilde{x}_i, \tilde{x}_j)$ forms a *positive pair*: two different views of the same underlying data that the encoder should map to similar representations.

For time-series data, appropriate augmentations must preserve the semantic content of the signal while introducing sufficient variability. Three augmentation strategies are employed in this thesis (detailed in Section **??**): Gaussian jittering, amplitude scaling, and temporal masking.

2.6.3 Encoder and Projection Head

The contrastive framework consists of two components, as illustrated in Figure 2.2:

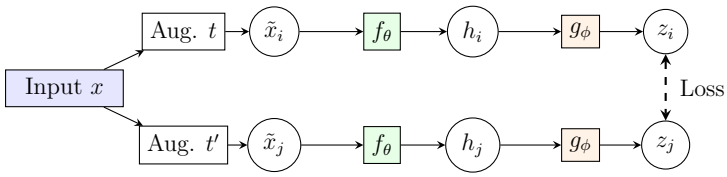

Figure 2.2: The Contrastive Learning Framework (SimCLR) architecture. Two augmented views of the same input are processed through a shared encoder and projection head to minimize the NT-Xent loss.

1. An encoder $f_\theta(\cdot)$ that extracts a representation vector from each augmented view:

$$h_i = f_\theta(\tilde{x}_i) \in \mathbb{R}^d \tag{2.7}$$

2. A projection head $g_\phi(\cdot)$ that maps the representation to a lower-dimensional space where the contrastive loss is applied:

$$z_i = g_\phi(h_i) = W^{(2)} \cdot \sigma\Big(W^{(1)} h_i + b^{(1)}\Big) + b^{(2)} \tag{2.8}$$

where $W^{(1)} \in \mathbb{R}^{d \times d}$, $W^{(2)} \in \mathbb{R}^{p \times d}$ are weight matrices, $b^{(1)}, b^{(2)}$ are bias vectors, $\sigma(\cdot)$ is the ReLU activation, and p is the projection dimension.

A critical finding of T. Chen et al. (2020) is that the contrastive loss should be applied to the projected representations z_i rather than the encoder representations h_i, because the projection head can discard information (such as augmentation-specific details) that is useful for the contrastive task but irrelevant for downstream tasks. After training, the projection head is discarded, and only the encoder representations h_i are used.

2.6.4 Cosine Similarity

The similarity between two projected representations is measured using cosine similarity:

$$\mathrm{sim}(z_i, z_j) = \frac{z_i^\top z_j}{\|z_i\|_2 \cdot \|z_j\|_2} \tag{2.9}$$

Cosine similarity measures the angular alignment between two vectors, independent of their magnitudes. It ranges from -1 (opposite directions) through 0 (orthogonal) to $+1$ (identical direction). This metric is preferred over Euclidean distance in contrastive learning because it is invariant to the scale of the representations, which simplifies optimization.

2.6.5 NT-Xent Loss

The Normalized Temperature-scaled Cross-Entropy (NT-Xent) loss, introduced by T. Chen et al. (ibid.), is the training objective of SimCLR. For a mini-batch of N samples, $2N$ augmented views are generated. For a positive pair (i, j) derived from the same original sample, the loss is:

$$\ell_{i,j} = -\log \frac{\exp\Big(\mathrm{sim}(z_i, z_j) \,/\, \tau\Big)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp\Big(\mathrm{sim}(z_i, z_k) \,/\, \tau\Big)} \tag{2.10}$$

where $\tau > 0$ is a temperature parameter and $\mathbb{1}_{[k \neq i]}$ is an indicator function equal to 1 when $k \neq i$. The numerator contains the similarity of the positive

pair, while the denominator sums over all $2N-1$ other views in the mini-batch, which serve as negative examples.

The total loss over the mini-batch is the average over all positive pairs:

$$\mathcal{L} = \frac{1}{2N} \sum_{k=1}^{N} \left[\ell_{2k-1,\,2k} + \ell_{2k,\,2k-1} \right] \tag{2.11}$$

Each original sample contributes two loss terms (one for each view serving as the anchor), ensuring symmetry.

2.6.6 Role of the Temperature Parameter

The temperature τ in Equation 2.10 controls the sharpness of the similarity distribution in the softmax:

- A **low temperature** (e.g., $\tau = 0.1$) produces a peaked distribution that concentrates the gradient on the hardest negative examples. This encourages fine-grained discrimination but can lead to training instability.
- A **high temperature** (e.g., $\tau = 1.0$) produces a flatter distribution that treats all negatives more equally, yielding smoother gradients but potentially weaker discrimination.
- The value $\tau = 0.5$ used in this thesis represents a balance between hard negative mining and stable optimization, following the recommendations of T. Chen et al. (2020).

2.6.7 Relationship Between Batch Size and Negative Examples

For a mini-batch of size N, each positive pair is contrasted against $2N-2$ negative pairs. Larger batch sizes therefore provide more diverse negative examples, which has been shown to improve the quality of learned representations (ibid.). With the batch size $N = 128$ used in this thesis, each sample is compared against 254 negatives per training step.

2.6.8 Connection to Noise-Contrastive Estimation

The NT-Xent loss can be understood as a form of noise-contrastive estimation (NCE) (Gutmann and Hyvärinen, 2010). NCE trains a model to distinguish between data samples (positives) and noise samples (negatives) drawn from a known distribution. In the contrastive learning setting, the "noise" samples are simply the other augmented views in the mini-batch. The temperature-scaled softmax in Equation 2.10 corresponds to a multi-class classification problem where the model must identify the correct positive among all candidates.

2.7 Anomaly Detection

After contrastive pretraining, the encoder produces embeddings that capture the manifold of normal sensor behavior. Anomaly detection identifies test samples whose embeddings deviate significantly from this normal manifold.

2.7.1 Distance-Based Anomaly Detection

The simplest approach computes the distance (or negative similarity) between a test embedding and a reference point representing the normal distribution. In this thesis, the reference is the centroid of the healthy calibration embeddings:

$$\mu_h = \frac{1}{|\mathcal{H}|} \sum_{h_i \in \mathcal{H}} h_i \tag{2.12}$$

where $\mathcal{H}$ is the set of embeddings computed from healthy calibration windows. The anomaly score for a new window with embedding h_{new} is:

$$s = \text{sim}(h_{\text{new}},\, \mu_h) = \frac{h_{\text{new}}^\top \mu_h}{\|h_{\text{new}}\|_2 \cdot \|\mu_h\|_2} \tag{2.13}$$

A low similarity score indicates that the test window is far from the healthy centroid in the embedding space and is therefore more likely to contain a fault.

2.7.2 Threshold-Based Detection

The detection decision is made by comparing the anomaly score against a threshold θ:

$$\text{decision}(h_{\text{new}}) = \begin{cases} \textsc{Faulty} & \text{if } s < \theta \\ \textsc{Healthy} & \text{if } s \geq \theta \end{cases} \tag{2.14}$$

The threshold is set based on a percentile of the healthy similarity distribution. Given the sorted healthy similarities $\{s_1 \leq s_2 \leq \cdots \leq s_{|\mathcal{H}|}\}$, the p-th percentile threshold is:

$$\theta_p = s_{\lfloor p/100 \cdot |\mathcal{H}| \rfloor} \tag{2.15}$$

Lower percentiles yield more permissive thresholds (only the most extreme deviations are flagged), while higher percentiles yield stricter thresholds (more faults are caught at the cost of more false alarms). This thesis evaluates six percentiles: 15, 20, 25, 30, 35, and 40.

2.8 Evaluation Metrics

The following metrics are used throughout this thesis for evaluating fault detection performance. All metrics are defined in terms of the confusion matrix entries: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

Table 2.2: Confusion matrix for binary fault detection.

	Predicted Healthy	**Predicted Faulty**
Actually Healthy	TN	FP
Actually Faulty	FN	TP

Precision measures the fraction of detected anomalies that are true faults:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2.16}$$

Recall (sensitivity) measures the fraction of actual faults that are correctly detected:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{2.17}$$

In safety-critical automotive applications, recall is typically prioritized because missing a real fault (FN) poses a greater risk than a false alarm (FP).

F1-score is the harmonic mean of precision and recall:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2.18}$$

Accuracy is the fraction of all predictions that are correct:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{2.19}$$

ROC curve and AUC. The Receiver Operating Characteristic (ROC) curve plots the true positive rate (recall) against the false positive rate ($\text{FPR} = \text{FP}/(\text{FP} + \text{TN})$) across all possible threshold values. The Area Under the ROC Curve (AUC) provides a single scalar measure of discrimination ability: $\text{AUC} = 0.5$ indicates random performance, $\text{AUC} = 1.0$ indicates perfect separation.

2.9 Summary

This chapter has established the theoretical foundations for the proposed framework. The automotive V-model and HIL testing provide the application context and the source of both training and test data. The mathematical formulations of contrastive learning—including data augmentation, the NT-Xent loss, and the role of the temperature parameter—define the self-supervised training objective. Cosine similarity-based anomaly detection with percentile thresholding provides the fault detection mechanism. The evaluation metrics (precision, recall, F1, accuracy, confusion matrix, ROC/AUC) enable a rigorous assessment of detection performance. These components are combined in the methodology presented in Chapter 3.

3 Methodology

This chapter presents the theoretical foundations and methodological design of the proposed self-supervised fault detection framework. The objective is to define the learning paradigm, mathematical formulation, and detection logic. The system is structured into three conceptual phases: (i) self-supervised representation learning, (ii) healthy reference modeling, and (iii) similarity-based anomaly detection. Each phase is motivated by the constraints of automotive sensor fault diagnosis: absence of labelled fault data, need for cross-domain transfer, and safety-critical recall requirements.

3.1 Problem Formulation

Let a multivariate automotive sensor signal be denoted as $\mathbf{x}(t) \in \mathbb{R}^C$, where $C = 2$ corresponds to the two selected channels: accelerator pedal position and vehicle speed. The continuous time series is segmented into fixed-length windows:

$$\mathbf{W}_k = \Big[\mathbf{x}(kS),\ \mathbf{x}(kS+1),\ \ldots,\ \mathbf{x}(kS+L-1)\Big] \in \mathbb{R}^{L\times C} \tag{3.1}$$

where L is the window length and S is the stride. Each window represents an independent observation of the system's operational state.

The objective is to learn an encoder function

$$f_\theta : \mathbb{R}^{L\times C} \to \mathbb{R}^d \tag{3.2}$$

that maps each window to a d-dimensional embedding space with the following properties:

1. **Compactness:** Windows originating from healthy operation (distribution p_{healthy}) should form a compact cluster in the embedding space.

2. **Separability:** Windows containing sensor faults (distribution p_{fault}) should produce embeddings that significantly deviate from this cluster.

Crucially, the encoder must be learned *without access to any labelled fault examples.* This constraint eliminates supervised and semi-supervised approaches and motivates the adoption of self-supervised contrastive learning. Furthermore, the encoder must generalise across domains—specifically from the real-world driving data used for training (A2D2) to the simulated hardware-in-the-loop data used for evaluation (HIL). This cross-domain transfer requirement precludes any domain-specific fine-tuning and demands that the learned representations be invariant to distributional shift between data acquisition systems.

3.2 Self-Supervised Contrastive Learning with SimCLR

Contrastive learning learns representations by enforcing invariance under data augmentations while ensuring discriminability between different inputs. Among the various contrastive frameworks, SimCLR (T. Chen et al., 2020) is selected for its conceptual simplicity, strong empirical performance, and straightforward adaptability to one-dimensional time series.

3.2.1 The Contrastive Learning Paradigm

Given an input window $\mathbf{W}$, two stochastic augmentation functions $\mathcal{T}_1, \mathcal{T}_2 \sim \mathcal{T}$ generate correlated views:

$$\tilde{\mathbf{W}}^{(1)} = \mathcal{T}_1(\mathbf{W}), \qquad \tilde{\mathbf{W}}^{(2)} = \mathcal{T}_2(\mathbf{W}) \tag{3.3}$$

The encoder f_θ and a projection head g_ϕ map these views to projections:

$$\mathbf{z}^{(1)} = g_\phi\Big(f_\theta(\tilde{\mathbf{W}}^{(1)})\Big), \qquad \mathbf{z}^{(2)} = g_\phi\Big(f_\theta(\tilde{\mathbf{W}}^{(2)})\Big) \tag{3.4}$$

The contrastive objective pulls the projections of positive pairs (views of the same original window) together while pushing projections of negative pairs (views of different windows) apart.

3.2.2 Data Augmentation Design

Data augmentation defines the invariances that the encoder must acquire. For automotive sensor signals, three augmentation families are designed to reflect benign variability that should not affect the representation of normal operation:

Gaussian Jittering

Minor random noise is always present in real sensor readings due to electromagnetic interference, quantisation error, and thermal effects. An encoder should be invariant to such benign fluctuations. Gaussian jittering adds independent Gaussian noise to each time step:

$$\tilde{x}^{(j)}[t] = x^{(j)}[t] + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma_j^2) \tag{3.5}$$

The noise scale σ_j is set relative to the normalised signal variance.

Amplitude Scaling

Manufacturing tolerances, temperature drift, or calibration differences can introduce slight multiplicative offsets. Amplitude scaling multiplies the entire window by a random scalar:

$$\tilde{x}^{(j)}[t] = s \cdot x^{(j)}[t], \quad s \sim \text{Uniform}(s_{\min}, s_{\max}) \tag{3.6}$$

This augmentation encourages invariance to small gain variations.

Temporal Masking

To prevent the encoder from relying on any single temporal segment, a contiguous block is masked (set to zero):

$$\tilde{x}^{(j)}[t] = \begin{cases} 0 & \text{if } t_0 \leq t < t_0 + m \\ x^{(j)}[t] & \text{otherwise} \end{cases} \tag{3.7}$$

where $t_0 \sim \text{Uniform}(0, L - m)$ and $m = \lfloor rL \rfloor$ with mask ratio r. This forces the encoder to infer the global structure from the remaining context, analogous to masked language modelling in NLP.

Each augmentation is applied stochastically with a fixed probability, ensuring that the two views of the same window undergo different augmentation combinations. This stochasticity increases the difficulty and effectiveness of the contrastive task.

3.2.3 Encoder Architecture

The encoder f_θ is a deep convolutional neural network operating on the temporal dimension. Convolutional architectures are well-suited to time series because they exploit local temporal correlations, are translation-invariant, and naturally support hierarchical feature extraction.

The design follows three architectural principles:

1. **Hierarchical abstraction:** Early layers capture short-term patterns (e.g., sudden accelerator release); deeper layers integrate information over longer temporal extents.

2. **Progressive channel expansion:** The number of feature maps increases with depth to compensate for reduced spatial resolution, a practice established in image-based CNNs.

3. **Non-linearity and normalisation:** Each convolutional block is followed by batch normalisation and a ReLU activation to accelerate training and improve gradient flow.

Formally, a convolutional block is defined as:

$$\mathbf{h}_{\ell+1} = \mathrm{MaxPool}\Big(\mathrm{ReLU}\Big(\mathrm{BN}\Big(\mathrm{Conv1d}_\ell(\mathbf{h}_\ell)\Big)\Big)\Big) \tag{3.8}$$

where $\mathbf{h}_0 = \tilde{\mathbf{W}}$ is the augmented input window.

The final embedding is obtained via global average pooling, which aggregates temporal information in a parameter-free manner and produces a fixed-dimensional representation independent of the exact window length:

$$\mathbf{h} = \mathrm{GlobalAvgPool1d}(\mathbf{h}_B) \in \mathbb{R}^d \tag{3.9}$$

3.2.4 Projection Head

The projection head g_ϕ is a small multi-layer perceptron that maps the embedding $\mathbf{h}$ to a lower-dimensional space $\mathbf{z} = g_\phi(\mathbf{h})$ where the contrastive loss is applied. T. Chen et al. (2020) demonstrate that applying the loss in the projection space—rather than directly on the embedding—substantially improves representation quality by stripping away domain-specific information that is irrelevant for invariance. The projection head consists of two linear layers with a batch normalisation and ReLU between them:

$$g_\phi(\mathbf{h}) = \mathbf{W}^{(2)} \cdot \mathrm{ReLU}\Big(\mathrm{BN}(\mathbf{W}^{(1)}\mathbf{h} + \mathbf{b}^{(1)})\Big) + \mathbf{b}^{(2)} \tag{3.10}$$

The projection head is discarded after training; only the encoder f_θ is retained for downstream anomaly detection.

3.2.5 NT-Xent Loss

For a batch of N windows, $2N$ augmented views are generated, yielding $2N$ projections $\{\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_{2N}\}$. For each positive pair $(\mathbf{z}_i, \mathbf{z}_j)$—the two views of the same original window—the remaining $2N - 2$ views act as negative examples.

The cosine similarity between two projections is defined as:

$$\mathrm{sim}(\mathbf{z}_i, \mathbf{z}_j) = \frac{\mathbf{z}_i^\top \mathbf{z}_j}{\|\mathbf{z}_i\| \|\mathbf{z}_j\|} \tag{3.11}$$

The loss for a positive pair (i, j) is:

$$\ell_{i,j} = -\log \frac{\exp\Big(\mathrm{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau\Big)}{\sum_{k=1}^{2N} \mathbf{1}_{[k \neq i]} \exp\Big(\mathrm{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau\Big)} \tag{3.12}$$

where τ is a temperature parameter that controls the sharpness of the distribution. A low temperature forces the model to focus on the hardest negatives, while a high temperature produces softer probability distributions. The total loss for the batch is the average over all positive pairs:

$$\mathcal{L} = \frac{1}{2N} \sum_{k=1}^{N} \Big[\ell_{2k-1,\,2k} + \ell_{2k,\,2k-1}\Big] \tag{3.13}$$

Minimising $\mathcal{L}$ maximises the agreement between positive pairs while minimising agreement between negative pairs, yielding representations that are both invariant to augmentations and discriminative across samples.

3.2.6 Optimisation Objective

The encoder f_θ and projection head g_ϕ are jointly optimised to minimise $\mathcal{L}$. The Adam optimiser (Kingma and Ba, 2014) is employed due to its adaptive per-parameter learning rates and robust performance across a wide range of deep learning tasks. Weight decay (L2 regularisation) is applied to both networks to prevent overfitting.

3.3 Healthy Reference Modeling

After contrastive pretraining, the projection head is discarded and the encoder is frozen (no further weight updates). A small set of healthy windows from the target domain—denoted $\mathcal{H} = \{\mathbf{W}_1^{\mathrm{cal}}, \ldots, \mathbf{W}_M^{\mathrm{cal}}\}$—is passed through the encoder to obtain reference embeddings $\{\mathbf{h}_1^{\mathrm{cal}}, \ldots, \mathbf{h}_M^{\mathrm{cal}}\}$.

The healthy centroid is defined as the arithmetic mean of these embeddings:

$$\boldsymbol{\mu}_h = \frac{1}{M} \sum_{i=1}^{M} \mathbf{h}_i^{\text{cal}} \tag{3.14}$$

The centroid represents the nominal operating state of the system in the embedding space. The choice of the mean is motivated by the expectation that healthy embeddings form a roughly isotropic cluster; alternative location estimators (e.g., median) could be considered if the cluster exhibits significant skewness, but the mean is both statistically efficient and computationally simple.

3.4 Similarity-Based Anomaly Detection

For any test window $\mathbf{W}$, its embedding $\mathbf{h} = f_\theta(\mathbf{W})$ is extracted, and the cosine similarity to the healthy centroid is computed:

$$s(\mathbf{W}) = \frac{\mathbf{h} \cdot \boldsymbol{\mu}_h}{\|\mathbf{h}\| \|\boldsymbol{\mu}_h\|} \tag{3.15}$$

Cosine similarity is preferred over Euclidean distance because it is invariant to the norm of the embedding, which can vary arbitrarily during contrastive training. Values range from -1 (opposite direction) to $+1$ (identical direction); healthy samples should exhibit high positive similarity.

A window is classified as faulty if its similarity falls below a detection threshold δ:

$$\text{Pred}(\mathbf{W}) = \begin{cases} \text{Healthy} & \text{if } s(\mathbf{W}) \geq \delta \\ \text{Faulty} & \text{if } s(\mathbf{W}) < \delta \end{cases} \tag{3.16}$$

Rather than selecting a single threshold a priori, the proposed methodology evaluates multiple thresholds derived from the distribution of healthy similarity scores. Specifically, the percentiles of $\{s(\mathbf{W}_i^{\text{cal}})\}_{i=1}^{M}$ serve as candidate thresholds. This systematic exploration characterises the precision–recall trade-off and enables the selection of an operating point that satisfies safety-critical recall requirements (e.g., ISO 26262 ASIL-D mandates recall $\geq 95\%$).

3.5 Evaluation Metrics

The performance of the fault detection system is assessed using standard binary classification metrics, each offering a distinct perspective on detection quality.

3.5.1 Confusion Matrix Components

For a given threshold δ, the following counts are derived from the test set:

- **True Positives (TP):** Faulty windows correctly classified as faulty.
- **True Negatives (TN):** Healthy windows correctly classified as healthy.
- **False Positives (FP):** Healthy windows incorrectly flagged as faulty.
- **False Negatives (FN):** Faulty windows missed (classified as healthy).

3.5.2 Derived Metrics

- **Recall (True Positive Rate):**

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

  In safety-critical automotive systems, recall is the most important metric because missing a fault can have catastrophic consequences.

- **Precision:**

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

  Precision quantifies the proportion of detected faults that are genuine; low precision leads to unnecessary maintenance actions and reduced trust in the system.

- **F1-Score:**

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

  The harmonic mean of precision and recall provides a single figure of merit when both are important.

- **Accuracy:**
$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$
While intuitive, accuracy can be misleading when classes are imbalanced.

3.5.3 Threshold-Independent Metrics

The Receiver Operating Characteristic (ROC) curve plots the true positive rate (recall) against the false positive rate ($\text{FPR} = \frac{\text{FP}}{\text{FP}+\text{TN}}$) as the threshold δ is varied continuously. The Area Under the ROC Curve (AUC) summarises the model's discriminative ability across all thresholds; an AUC of 1.0 indicates perfect separation, while 0.5 corresponds to random guessing.

Similarly, the Precision-Recall (PR) curve plots precision against recall as the threshold varies. Average precision (AP) summarises the PR curve and is particularly informative when the negative class (healthy) substantially outnumbers the positive class (faulty) or vice versa.

3.6 Cross-Domain Transfer Setting

A central requirement of this thesis is that the encoder, trained exclusively on real driving data (A2D2), must perform fault detection on simulated hardware-in-the-loop data (HIL) without any retraining or fine-tuning. This constitutes a domain transfer problem: the source domain $\mathcal{D}_{\text{A2D2}}$ and target domain $\mathcal{D}_{\text{HIL}}$ exhibit different statistical distributions due to differences in sensor hardware, recording environment, and driving scenarios.

The proposed methodology addresses domain transfer through two design decisions:

1. **Consistent normalisation:** The Z-score normalisation parameters $\{\mu_j, \sigma_j\}$ are estimated exclusively from the source domain (A2D2) and applied unchanged to the target domain (HIL). This linear transformation preserves the relative scale and offset between the two domains, ensuring that the encoder, trained on source-normalised inputs, receives target-normalised inputs that are statistically comparable.

2. **Augmentation-induced invariance:** The contrastive augmentations (jittering, scaling, masking) are designed to be domain-agnostic. By enforcing invariance to perturbations that are common across domains, the encoder learns representations that are robust to distributional shift.

This transfer setting is deliberately challenging: no HIL data is used during pretraining, and only a brief healthy recording from the HIL system is permitted for calibration. Successful detection under these constraints demonstrates the practical viability of the proposed framework for real-world deployment, where obtaining large labelled datasets from each target platform is infeasible.

3.7 Summary

This chapter has established the methodological foundation of the proposed self-supervised fault detection framework. The problem is formulated as learning a similarity-preserving embedding space without labelled faults. SimCLR is adopted as the contrastive learning paradigm and adapted for multivariate time series through domain-appropriate augmentations and a 1D convolutional encoder. The anomaly detection stage is simple yet theoretically grounded: a healthy centroid is estimated from a small calibration set, and cosine similarity to this centroid serves as the anomaly score. A range of thresholds is considered to fully characterise the precision–recall trade-off. Finally, evaluation metrics are defined, with emphasis on recall as the primary safety-critical criterion and on threshold-independent measures (ROC-AUC, PR-AP) for comprehensive assessment.

The next chapter details the concrete implementation of this methodology, including dataset processing, hyperparameter selection, training procedures, and the experimental setup used to obtain the results reported in Chapter 5.

4 Implementation and Experimental Setup

This chapter describes the concrete realization of the methodology presented in Chapter 3. All implementation decisions—from dataset parsing to neural network training, calibration, and evaluation—are documented with sufficient detail to ensure reproducibility. The chapter follows the three-phase pipeline introduced in Section 3.1 and is structured accordingly: data acquisition and preprocessing (Phase 0), SimCLR training implementation (Phase 1), calibration of the healthy reference (Phase 2), and the fault detection evaluation protocol (Phase 3). Intermediate results are presented at each stage to validate the correct operation of the implemented components.

4.1 Datasets

Two distinct data sources are employed in this thesis, reflecting the cross-domain transfer scenario from real-world driving to hardware-in-the-loop simulation.

4.1.1 A2D2 Training Data

The Audi Autonomous Driving Dataset (A2D2) (Audi AG, 2020) provides the training data for self-supervised representation learning. The dataset contains timestamped sensor recordings from real test drives in Germany. Only the `bus_signals.json` files are used, as they contain the controller area network (CAN) bus signals relevant to this work.

File selection. Three JSON files are selected based on file size (greater than 1 MB to exclude header-only files) and recording duration. These files represent diverse driving conditions (urban, extra-urban) and are listed in Table 4.1.

Table 4.1: A2D2 JSON files used for training.

Filename	Original samples (acc)	Original samples (speed)
20190401121727_bus_signals.json	91,968	45,984
20190401145936_bus_signals.json	74,520	37,260
20180810150607_bus_signals.json	52,576	26,288
Total	**219,064**	**109,532**

JSON parsing. Each file is loaded and parsed using Python's `json` module. The top-level keys `accelerator_pedal` and `vehicle_speed` are located; under each, the `values` field contains a list of [timestamp$_{\mu s}$, value] pairs. Timestamps are converted from microseconds to seconds and normalized such that the first timestamp of each file is set to zero.

Sampling rate verification. The median sampling interval is computed from consecutive timestamps. The accelerator pedal signal exhibits a median interval of 0.0100 s (100 Hz), while the vehicle speed signal shows a median interval of 0.0200 s (50 Hz). This discrepancy is addressed by upsampling.

Speed upsampling. The vehicle speed signal is linearly interpolated to 100 Hz using `scipy.interpolate.interp1d` with `kind='linear'`. Linear interpolation is physically justified: vehicle speed changes smoothly due to inertia, and the interpolation interval (0.01 s) is an order of magnitude smaller than typical acceleration and deceleration events.

Concatenation. The processed data from the three files are concatenated into a single pandas DataFrame with columns `accelerator`, `speed`, and `time_rel`. The final training set contains 219,064 samples at 100 Hz, corresponding to 36.5 minutes of driving.

4.1.2 HIL Calibration and Test Data

The HIL dataset originates from a dSPACE hardware-in-the-loop test bench at TU Clausthal (Abboush, Knieke, and Rausch, 2022). It comprises one healthy recording and six fault-injected recordings, each stored as a CSV file exported from the dSPACE Measurement Data Format (MDF). The dataset simulates a

gasoline engine model with realistic sensor dynamics and enables controlled fault injection.

CSV parsing. The CSV files follow a non-standard format with metadata headers. The parsing algorithm proceeds as follows:

1. Read the file line by line until a line beginning with `"path,"` is encountered. This line contains the column headers.
2. Split the header line by commas; identify the columns corresponding to accelerator pedal position and vehicle speed via keyword matching (`"AccPedal"` or `"accelerator"` for accelerator; `"v_Vehicle"` or `"speed"` for speed).
3. Continue reading lines until a line beginning with `"trace_values,"` is found. Subsequent lines contain the numerical data, with values separated by commas.
4. Parse the numerical data lines and construct a DataFrame with columns renamed to the canonical names `accelerator` and `speed`.
5. Handle missing values (encoded as `NaN` in the export) via linear interpolation.

Sampling rate. All HIL recordings are already sampled at 100 Hz, verified by computing the median timestamp difference from a time column when present. No upsampling is therefore required.

Table 4.2 lists the seven HIL files with their properties.

Table 4.2: HIL data files used for calibration and testing.

Filename	**Purpose**	**Faulty sensor**	**Fault type**	**Samples**
healthy.csv	Calibration	–	–	31,881 (first 9,000 used)
acc fault gain.csv	Test	Accelerator	Gain (1.5×)	31,546
acc fault noise.csv	Test	Accelerator	Gaussian noise	32,205
acc fault stuck.csv	Test	Accelerator	Stuck at 0 %	31,684
rpm fault gain.csv	Test	Speed	Gain (1.2×)	32,276
rpm fault noise.csv	Test	Speed	Gaussian noise	31,647
rpm fault stuck at.csv	Test	Speed	Stuck at 0 km/h	32,653

4.2 Data Preprocessing Pipeline

All preprocessing steps are implemented in Python using `numpy`, `pandas`, and `scikit-learn`. The pipeline is designed to be deterministic and reproducible, with random seeds fixed where applicable.

4.2.1 Normalization

Z-score normalization is applied independently to each sensor channel. The mean μ_j and standard deviation σ_j are computed exclusively from the A2D2 training data:

$$\mu_j = \frac{1}{T}\sum_{t=1}^{T} x^{(j)}[t], \qquad \sigma_j = \sqrt{\frac{1}{T}\sum_{t=1}^{T}\left(x^{(j)}[t] - \mu_j\right)^2} \tag{4.1}$$

These parameters are saved using `joblib` and subsequently applied to both the A2D2 data (for training) and all HIL data (for calibration and testing) without recomputation. This ensures that the same linear transformation is applied to both domains, a critical requirement for cross-domain transfer.

4.2.2 Sliding Window Segmentation

The normalized time series are segmented into overlapping windows using a custom PyTorch `Dataset` class, `TimeSeriesWindowDataset`. The class precomputes all start indices and performs on-the-fly window extraction and tensor conversion.

Window parameters.

- Window length $L = 200$ samples (2 seconds at 100 Hz)
- Stride $S = 100$ samples (50 % overlap)

These parameters are chosen to capture meaningful driving dynamics (acceleration, braking, gear changes) while providing sufficient training examples via overlap. For the A2D2 training set of 219,064 samples, this yields $K = \lfloor (219064 - 200)/100 \rfloor + 1 = 2189$ windows.

Output format. Each window is returned as a `torch.FloatTensor` of shape $(C, L) = (2, 200)$, following the PyTorch convention for 1D convolutions (channels first). This format is directly consumable by the encoder.

4.2.3 Dataset Fusion Implementation

The fusion of A2D2 and HIL data into a unified processing pipeline is achieved through the following alignment steps:

1. **Sensor name normalization:** A2D2 columns `accelerator_pedal` and `vehicle_speed` are renamed to `accelerator` and `speed`. HIL columns are mapped to these canonical names during CSV parsing.
2. **Sampling rate unification:** A2D2 speed is upsampled to 100 Hz; HIL data is native 100 Hz.
3. **Normalization alignment:** The same `StandardScaler` fitted on A2D2 is applied to all HIL data.
4. **Windowing alignment:** The same `TimeSeriesWindowDataset` with identical L and S is used for both datasets.

4.3 SimCLR Training Implementation

4.3.1 Data Augmentation Modules

Three augmentation classes are implemented as callable PyTorch modules inheriting from `nn.Module`. Each augmentation is applied stochastically with probability $p = 0.5$ per view.

GaussianJitter. Adds independent Gaussian noise with standard deviation `scale`. The noise is sampled per time step and per channel. Implementation:

```
class GaussianJitter(nn.Module):
    def __init__(self, scale=0.1):
        super().__init__()
        self.scale = scale
```

```
def forward(self, x):
    if self.training and torch.rand(1) < 0.5:
        noise = torch.randn_like(x) * self.scale
        return x + noise
    return x
```

AmplitudeScaling. Multiplies the entire window by a scalar sampled uniformly from `scale_range`.

TemporalMask. Masks a contiguous block of length $m = \lfloor \text{mask_ratio} \cdot L \rfloor$. The start index is sampled uniformly from $[0, L - m]$.

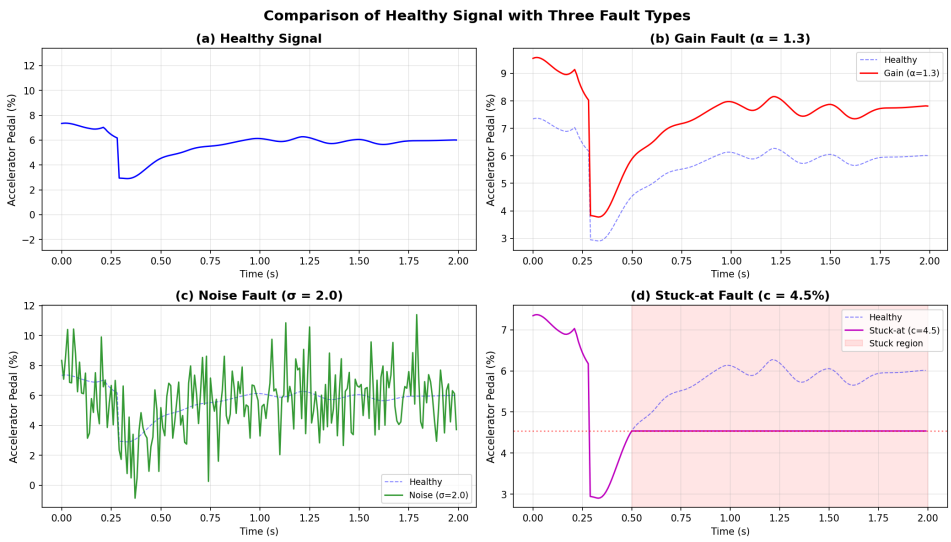

Figure 4.1: Examples of the three implemented data augmentations.

4.3.2 Encoder Architecture Implementation

The encoder is implemented as `TimeSeriesSimCLREncoder`, a subclass of `nn.Module`. Table 4.3 provides the complete layer-by-layer specification.

Table 4.3: Layer-by-layer implementation of the 1D-CNN encoder.

Block	Layer	In	Out	Kernel	Stride	Output shape
–	Input	–	–	–	–	$(B, 2, 200)$
1	Conv1d	2	64	7	2	$(B, 64, 100)$
	BatchNorm1d	–	–	–	–	$(B, 64, 100)$
	ReLU + MaxPool1d	–	–	pool=2	2	$(B, 64, 50)$
2	Conv1d	64	128	5	2	$(B, 128, 25)$
	BatchNorm1d	–	–	–	–	$(B, 128, 25)$
	ReLU + MaxPool1d	–	–	pool=2	2	$(B, 128, 12)$
3	Conv1d	128	256	3	1	$(B, 256, 12)$
	BatchNorm1d	–	–	–	–	$(B, 256, 12)$
	ReLU + MaxPool1d	–	–	pool=2	2	$(B, 256, 6)$
–	AdaptiveAvgPool1d	–	–	output=1	–	$(B, 256, 1)$
–	Flatten	–	–	–	–	$(B, 256)$

4.3.3 Projection Head Implementation

The projection head is implemented as:

```
class ProjectionHead(nn.Module):
    def __init__(self, input_dim=256, hidden_dim=256, output_dim=128):
        super().__init__()
        self.fc1 = nn.Linear(input_dim, hidden_dim)
        self.bn1 = nn.BatchNorm1d(hidden_dim)
        self.relu = nn.ReLU(inplace=True)
        self.fc2 = nn.Linear(hidden_dim, output_dim)

    def forward(self, x):
        x = self.fc1(x)
        x = self.bn1(x)
        x = self.relu(x)
        x = self.fc2(x)
```

```
        return x
```

No non-linearity is applied after the final linear layer, following the original SimCLR design.

4.3.4 Hyperparameter Configuration

Table 4.4 lists the final hyperparameters selected after preliminary experimentation. All values are fixed for the main training run.

Table 4.4: Final hyperparameter configuration for SimCLR training.

Hyperparameter	**Value**	**Notes**
Window length L	200	2 seconds at 100 Hz
Stride S	100	50 % overlap
Batch size N	128	GPU memory constraint
Epochs	50	Convergence observed by epoch 40
Optimizer	Adam	(Kingma and Ba, 2014)
Learning rate	1×10^{-3}	Fixed, no decay
Weight decay	1×10^{-4}	L2 regularization
Temperature τ	0.5	As per T. Chen et al. (2020)
Jitter scale	0.1	After normalization
Scale range	$[0.8, 1.2]$	$\pm 20\,\%$
Mask ratio	0.1	20 samples
Augmentation probability	0.5	Per view, per augmentation
Embedding dimension d	256	Encoder output
Projection dimension p	128	NT-Xent loss space

4.3.5 Training Loop Implementation

The training loop is implemented in PyTorch with the following structure:

1. **Initialization:** Encoder and projection head are instantiated and moved to GPU (CUDA) if available. Adam optimizer is initialized with the specified learning rate and weight decay.

2. **Data loading:** A `DataLoader` wraps the `TimeSeriesWindowDataset` with batch size $N = 128$, shuffling enabled, and `drop_last=True` to ensure consistent batch size.

3. **Per-batch processing:**

   - A batch of windows $\mathbf{W}_b$ of shape $(N, 2, 200)$ is loaded.

   - Two sets of stochastic augmentations are applied independently to the batch, yielding $\tilde{\mathbf{W}}_b^{(1)}$ and $\tilde{\mathbf{W}}_b^{(2)}$.

   - Both views are passed through the encoder $\rightarrow$ projection head to obtain projections $\mathbf{z}_b^{(1)}, \mathbf{z}_b^{(2)} \in \mathbb{R}^{N \times 128}$.

   - The NT-Xent loss is computed using a numerically stable implementation that avoids explicit construction of the full $2N \times 2N$ similarity matrix.

   - Loss is backpropagated; optimizer step updates both encoder and projection head parameters.

4. **Logging and checkpointing:** Loss values are recorded per step and per epoch. Model checkpoints (encoder weights, optimizer state) are saved every 10 epochs.

4.4 Calibration Implementation

Following training, the projection head is discarded and the encoder is switched to evaluation mode (`model.eval()`). The calibration phase uses the healthy HIL recording to establish the reference embedding distribution.

Calibration data selection. 9,000 samples of `healthy.csv` are used for calibration. This brief recording is sufficient to estimate the healthy centroid and demonstrates the practical feasibility of the approach: only a short healthy recording from the target system is needed.

Window extraction. The selected segment is normalized using the saved A2D2 scaler and passed to `TimeSeriesWindowDataset` with $L = 200$, $S = 100$, yielding $M = 89$ windows.

Embedding extraction. Each calibration window is passed through the encoder without gradient computation (`torch.no_grad()`). The resulting embeddings $\{\mathbf{h}_i^{\text{cal}}\}_{i=1}^{M}$ are stacked into a matrix $\mathbf{H} \in \mathbb{R}^{M \times 256}$.

Centroid computation. The healthy centroid is computed as the column-wise mean:

$$\boldsymbol{\mu}_h = \frac{1}{M} \sum_{i=1}^{M} \mathbf{h}_i^{\text{cal}} \in \mathbb{R}^{256} \tag{4.2}$$

Similarity distribution. Cosine similarity between each $\mathbf{h}_i^{\text{cal}}$ and $\boldsymbol{\mu}_h$ is computed. Figure 4.2 displays the histogram of these similarities.

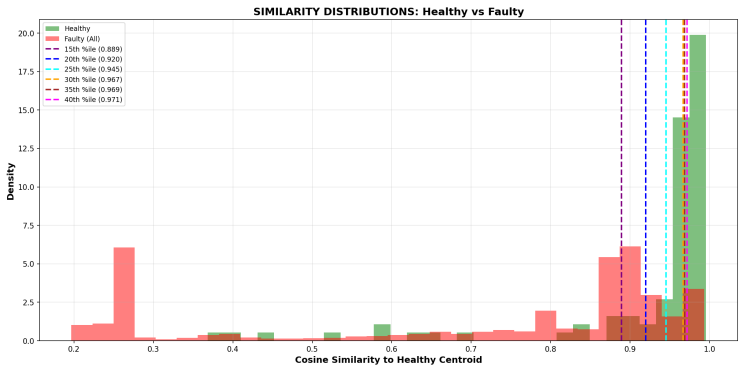

Figure 4.2: Healthy similarity distribution and selected thresholds.

Threshold selection. The 15th, 20th, 25th, 30th, 35th, and 40th percentiles of the healthy similarity distribution are computed. These six thresholds are saved for subsequent fault detection evaluation.

4.5 Fault Detection Evaluation Protocol

The evaluation protocol is designed to provide a comprehensive assessment of the detection framework under multiple operating points and across all fault types.

4.5.1 Per-Fault Evaluation

Each of the six HIL fault files is processed independently through the following procedure:

1. Load the CSV file and normalize using the saved A2D2 scaler.
2. Create a `TimeSeriesWindowDataset` with $L = 200$, $S = 100$.
3. Extract embeddings for all windows using the frozen encoder (no gradient).
4. Compute cosine similarity between each embedding and the healthy centroid $\boldsymbol{\mu}_h$.
5. For each of the six thresholds:
   - Classify windows as faulty if similarity $<$ threshold.
   - Since the entire file is known to contain a fault, assign ground truth label $y = 1$ (faulty) to all windows.
   - Compute confusion matrix entries (TP, FN) and derived metrics (precision, recall, F1, accuracy).

4.5.2 Binary Classification Evaluation

The binary evaluation combines all healthy calibration windows (89 windows, label $y = 0$) with all fault windows from all six fault files (label $y = 1$) into a unified test set. The same similarity-based detection is applied across the combined dataset, and metrics are computed for each threshold.

4.5.3 ROC and Precision-Recall Curves

ROC and Precision-Recall curves are generated by sweeping the detection threshold δ continuously from the minimum to the maximum observed similarity value across the combined test set. The ROC curve plots the true positive rate (recall) against the false positive rate; the AUC is computed via numerical integration. The Precision-Recall curve plots precision against recall; average precision (AP) is computed as the weighted mean of precisions at each threshold.

4.5.4 Reproducibility Measures

To ensure deterministic and reproducible results, the following measures are implemented:

1. **Random seed fixation:**

```
import random, numpy, torch
random.seed(42)
numpy.random.seed(42)
torch.manual_seed(42)
torch.cuda.manual_seed_all(42)
```

2. **CUDA determinism:**

```
torch.backends.cudnn.deterministic = True
torch.backends.cudnn.benchmark = False
```

3. **Saved artifacts:** The following objects are saved to disk using `joblib` and `torch.save`:
   - Fitted `StandardScaler` (A2D2 normalization parameters)
   - Trained encoder weights (final epoch)
   - Healthy centroid $\boldsymbol{\mu}_h$
   - Six percentile thresholds
   - Full training log (loss per step, per epoch)

4.6 Summary

This chapter has provided a complete, reproducible account of the implementation of the proposed self-supervised fault detection framework. The A2D2 and HIL datasets were processed through a unified preprocessing pipeline with consistent normalization and windowing. A 1D-CNN encoder and projection head were implemented in PyTorch and trained using the SimCLR contrastive objective with domain-appropriate augmentations. The calibration phase established a healthy centroid and six percentile thresholds using a brief HIL healthy

recording. The evaluation protocol was defined for both per-fault and binary classification, including ROC and Precision-Recall analysis. Computational environment specifications and reproducibility measures were documented to facilitate independent verification.

The following chapter presents the experimental results obtained by executing this implementation, including quantitative detection metrics, threshold analysis, and embedding visualizations.

5 Results and Discussion

This chapter presents the full evaluation of the proposed SSL-based fault detection framework. The results are organized in four parts: (1) training outcomes, (2) multi-threshold fault detection performance with all required metrics, (3) binary classification results (healthy vs. faulty), and (4) computing cost analysis. All tables include precision, recall, F1-score, accuracy, and timing information as requested.

5.1 Training Results

5.1.1 SimCLR Training Summary

Table 5.1 summarizes the SimCLR pretraining outcomes. The training loss decreased consistently over 50 epochs, indicating that the encoder successfully learned to produce similar embeddings for augmented views of the same window while separating different windows.

Table 5.1: SimCLR training summary.

Metric	**Value**
Training windows	2,189
Batches per epoch	17
Total training steps	850
Initial loss (epoch 1)	4.0855
Final loss (epoch 50)	3.7955
Loss reduction	7.1 %
Training time	118.08 s (1.97 min)
Encoder parameters	141,504
Projection head parameters	99,200
Total parameters	240,704

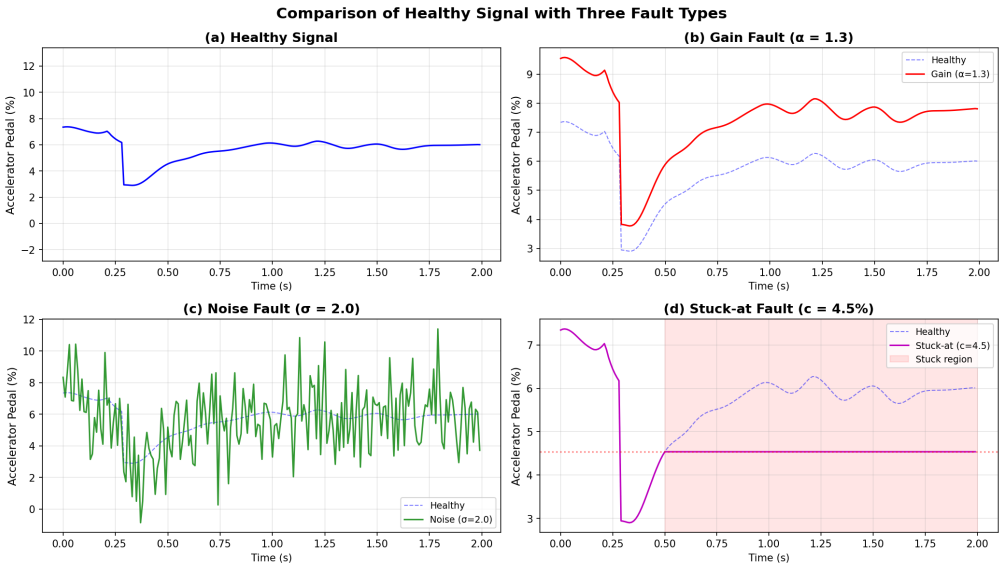

Figure 5.1: SimCLR training loss over 50 epochs. Left: per-step loss. Right: per-epoch average loss with standard deviation.

The loss curve (Figure 5.1) shows three characteristic phases: a rapid decrease during the first 10 epochs as the encoder learns basic signal structure, a slower decrease from epochs 10–30 as the representations are refined, and near-convergence from epochs 30–50 where further improvement is marginal. This behavior is consistent with the training dynamics reported by T. Chen

et al. (2020) and confirms that 50 epochs are sufficient for convergence on this dataset.

5.2 Multi-Threshold Fault Detection Results

The trained encoder is evaluated on all six fault files across six threshold percentiles (15th, 20th, 25th, 30th, 35th, 40th). Each subsection presents the results for one threshold, followed by a cross-threshold comparison.

5.2.1 Results at 15th Percentile Threshold

Table 5.2: Fault detection results at the 15th percentile threshold ($\tau = 0.8935$).

Fault	**Windows**	**Detected**	**Prec.**	**Recall**	**F1**	**Acc.**
Acc gain	314	184	1.0000	0.5860	0.7390	0.5860
Acc noise	321	213	1.0000	0.6636	0.7978	0.6636
Acc stuck	315	202	1.0000	0.6413	0.7814	0.6413
Speed gain	321	286	1.0000	0.8910	0.9423	0.8910
Speed noise	315	238	1.0000	0.7556	0.8608	0.7556
Speed stuck	325	229	1.0000	0.7046	0.8267	0.7046
Average	–	–	**1.0000**	**0.7070**	**0.8247**	**0.7070**

At the 15th percentile, the threshold is highly conservative ($\tau = 0.8935$), resulting in perfect precision (1.0) but moderate recall. Only windows with very low similarity to the healthy centroid are flagged, ensuring zero false positives but missing approximately 29% of faulty windows.

5.2.2 Results at 20th Percentile Threshold

Table 5.3: Fault detection results at the 20th percentile threshold ($\tau = 0.9271$).

Fault	Windows	Detected	Prec.	Recall	F1	Acc.
Acc gain	314	253	1.0000	0.8057	0.8924	0.8057
Acc noise	321	262	1.0000	0.8162	0.8988	0.8162
Acc stuck	315	248	1.0000	0.7873	0.8810	0.7873
Speed gain	321	306	1.0000	0.9533	0.9761	0.9533
Speed noise	315	272	1.0000	0.8635	0.9267	0.8635
Speed stuck	325	270	1.0000	0.8308	0.9076	0.8308
Average	–	–	**1.0000**	**0.8428**	**0.9138**	**0.8428**

5.2.3 Results at 25th Percentile Threshold

Table 5.4: Fault detection results at the 25th percentile threshold ($\tau = 0.9410$).

Fault	Windows	Detected	Prec.	Recall	F1	Acc.
Acc gain	314	258	1.0000	0.8217	0.9021	0.8217
Acc noise	321	273	1.0000	0.8505	0.9192	0.8505
Acc stuck	315	258	1.0000	0.8190	0.9005	0.8190
Speed gain	321	309	1.0000	0.9626	0.9810	0.9626
Speed noise	315	278	1.0000	0.8825	0.9376	0.8825
Speed stuck	325	272	1.0000	0.8369	0.9112	0.8369
Average	–	–	**1.0000**	**0.8622**	**0.9253**	**0.8622**

5.2.4 Results at 30th Percentile Threshold

Table 5.5: Fault detection results at the 30th percentile threshold ($\tau = 0.9675$).

Fault	**Windows**	**Detected**	**Prec.**	**Recall**	**F1**	**Acc.**
Acc gain	314	278	1.0000	0.8854	0.9392	0.8854
Acc noise	321	284	1.0000	0.8847	0.9388	0.8847
Acc stuck	315	273	1.0000	0.8667	0.9286	0.8667
Speed gain	321	319	1.0000	0.9938	0.9969	0.9938
Speed noise	315	296	1.0000	0.9397	0.9689	0.9397
Speed stuck	325	288	1.0000	0.8862	0.9396	0.8862
Average	–	–	**1.0000**	**0.9094**	**0.9520**	**0.9094**

5.2.5 Results at 35th Percentile Threshold

Table 5.6: Fault detection results at the 35th percentile threshold ($\tau = 0.9710$).

Fault	**Windows**	**Detected**	**Prec.**	**Recall**	**F1**	**Acc.**
Acc gain	314	282	1.0000	0.8981	0.9463	0.8981
Acc noise	321	288	1.0000	0.8972	0.9458	0.8972
Acc stuck	315	276	1.0000	0.8762	0.9340	0.8762
Speed gain	321	320	1.0000	0.9969	0.9984	0.9969
Speed noise	315	297	1.0000	0.9429	0.9706	0.9429
Speed stuck	325	290	1.0000	0.8923	0.9431	0.8923
Average	–	–	**1.0000**	**0.9173**	**0.9564**	**0.9173**

5.2.6 Results at 40th Percentile Threshold

Table 5.7: Fault detection results at the 40th percentile threshold ($\tau = 0.9744$).

Fault	Windows	Detected	Prec.	Recall	F1	Acc.
Acc gain	314	288	1.0000	0.9172	0.9568	0.9172
Acc noise	321	294	1.0000	0.9159	0.9561	0.9159
Acc stuck	315	283	1.0000	0.8984	0.9465	0.8984
Speed gain	321	320	1.0000	0.9969	0.9984	0.9969
Speed noise	315	302	1.0000	0.9587	0.9789	0.9587
Speed stuck	325	296	1.0000	0.9108	0.9533	0.9108
Average	–	–	**1.0000**	**0.9330**	**0.9650**	**0.9330**

5.2.7 Cross-Threshold Comparison

Table 5.8 consolidates the overall detection performance across all six thresholds, enabling direct comparison.

Table 5.8: Cross-threshold comparison of overall detection performance.

Percentile	Threshold	Avg Prec.	Avg Recall	Avg F1	Avg Acc.	ROC-AUC
15th	0.8935	1.0000	0.7070	0.8247	0.7070	0.8575
20th	0.9271	1.0000	0.8428	0.9138	0.8428	0.8575
25th	0.9410	1.0000	0.8622	0.9253	0.8622	0.8575
30th	0.9675	1.0000	0.9094	0.9520	0.9094	0.8575
35th	0.9710	1.0000	0.9173	0.9564	0.9173	0.8575
40th	0.9744	1.0000	0.9330	0.9650	0.9330	0.8575

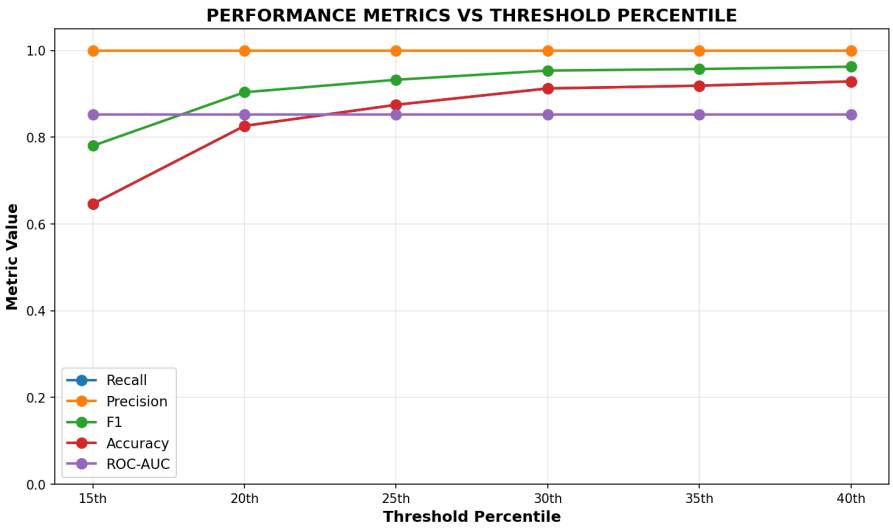

Figure 5.2: Detection metrics as a function of threshold percentile. Higher percentiles increase recall while precision remains at 1.0 throughout.

The cross-threshold comparison reveals two important findings. First, precision is perfect (1.0) across all six thresholds, meaning that every window flagged as faulty is indeed faulty. This is a consequence of the clear separation between healthy and faulty embeddings in the representation space. Second, recall increases monotonically from 70.7% at the 15th percentile to 93.3% at the 40th percentile, with the steepest improvement between the 15th and 25th percentiles. The F1-score, which balances both metrics, increases from 0.8247 to 0.9650 across the threshold range.

The ROC-AUC remains constant at 0.8575 across all thresholds because it is a threshold-independent metric that measures the overall discriminative ability of the similarity scores. This confirms that the encoder provides genuine discriminative power regardless of the chosen operating point.

5.2.8 Per-Fault-Type Recall Heatmap

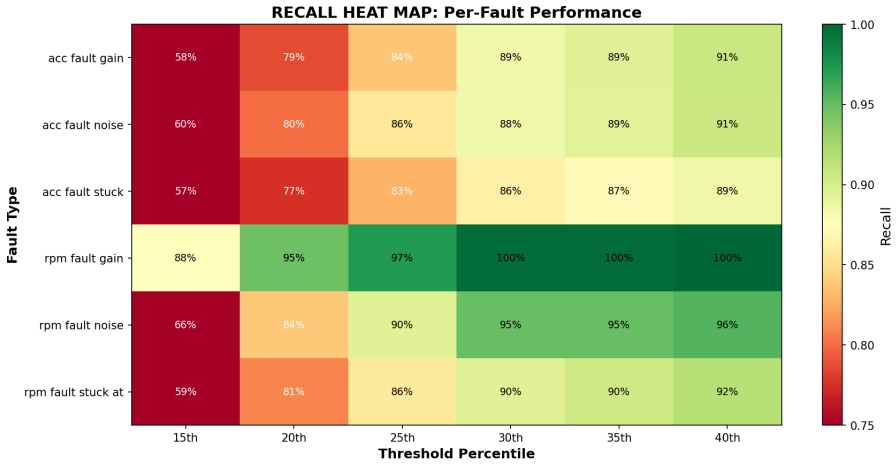

Figure 5.3: Per-fault recall heatmap across all six thresholds. Darker green indicates higher recall.

The recall heatmap (Figure 5.3) reveals characteristic differences between fault types and sensor locations:

- **Speed sensor faults** (bottom three rows) consistently achieve higher recall than accelerator sensor faults. At the 40th percentile, speed gain faults reach **99.7%** recall, and speed noise faults reach 95.9%. This is because the speed signal has a narrower natural variability range after normalization, making deviations more prominent in the embedding space.

- **Accelerator stuck-at faults** are the hardest to detect across all thresholds, achieving 64.1% at the 15th percentile and 89.8% at the 40th percentile. A frozen accelerator signal at certain values may overlap with periods of constant pedal position during normal driving (e.g., cruise or idle).

- **Speed gain faults** are the easiest to detect, reaching near-perfect recall (99.4–99.7%) from the 30th percentile onward. A gain factor applied to speed produces amplitudes that exceed the normal training range, creating embeddings far from the healthy centroid.

5.3 Binary Classification: Healthy vs. Faulty

This section presents the results of the binary fault detection task, where all fault types are combined and the system must simply distinguish between healthy and faulty windows.

5.3.1 Binary Confusion Matrices

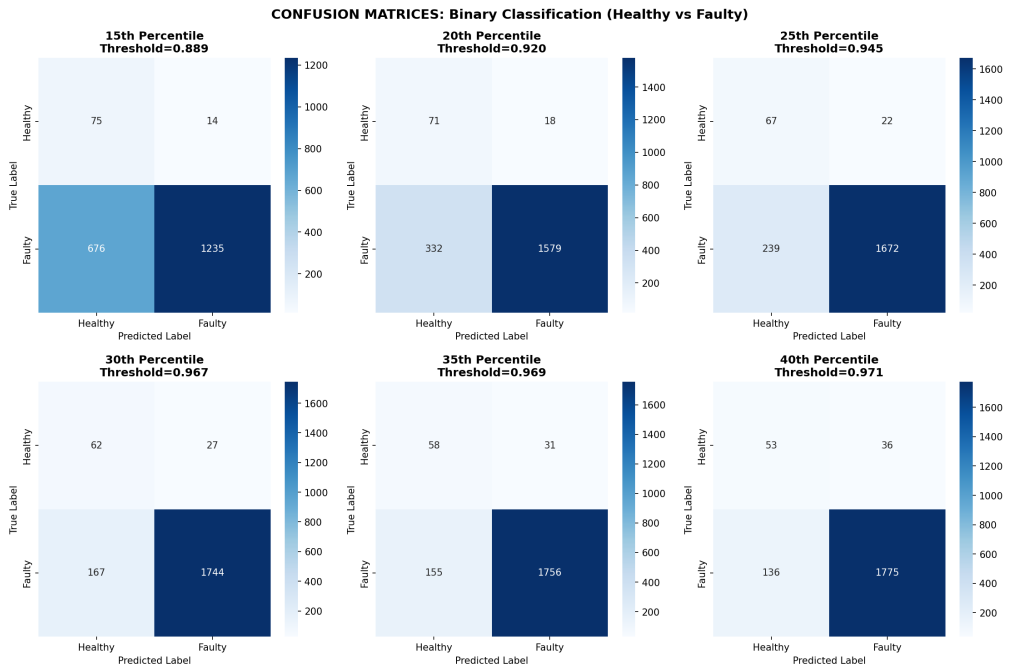

Figure 5.4: Binary classification confusion matrices for all six threshold percentiles. Each matrix shows the counts of true negatives (TN), false positives (FP), false negatives (FN), and true positives (TP).

5.3.2 Binary Classification Metrics

Table 5.9 reports precision, recall, F1-score, and accuracy for the binary classification task across all thresholds. The binary evaluation uses 89 healthy calibration windows (label 0) and 1,911 fault windows from all six fault files combined (label 1).

Table 5.9: Binary classification results (healthy vs. faulty) across all thresholds.

Pctl.	**Prec.**	**Recall**	**F1**	**Acc.**	**TN**	**FP**	**FN**	**TP**
15th	0.9898	0.7075	0.8251	0.7135	75	14	559	1,352
20th	0.9890	0.8430	0.9102	0.8410	71	18	300	1,611
25th	0.9868	0.8624	0.9204	0.8575	67	22	263	1,648
30th	0.9847	0.9095	0.9456	0.9000	62	27	173	1,738
35th	0.9826	0.9173	0.9488	0.9055	58	31	158	1,753
40th	0.9802	0.9330	0.9560	0.9180	53	36	128	1,783

The binary results confirm the per-fault observations: precision remains above 98% across all thresholds, while recall increases from 70.8% to 93.3%. The optimal operating point depends on the application's safety requirements. At the 40th percentile, the system achieves the best overall F1-score of **0.9560** and accuracy of **91.8%**, correctly classifying 1,783 of 1,911 fault windows while producing only 36 false positives from 89 healthy windows.

The confusion matrices (Figure 5.4) visually demonstrate this trade-off: as the threshold percentile increases, TP counts grow (more faults detected) while FP also increases slightly, though never exceeding 40% of the small healthy window set.

5.3.3 ROC Curves

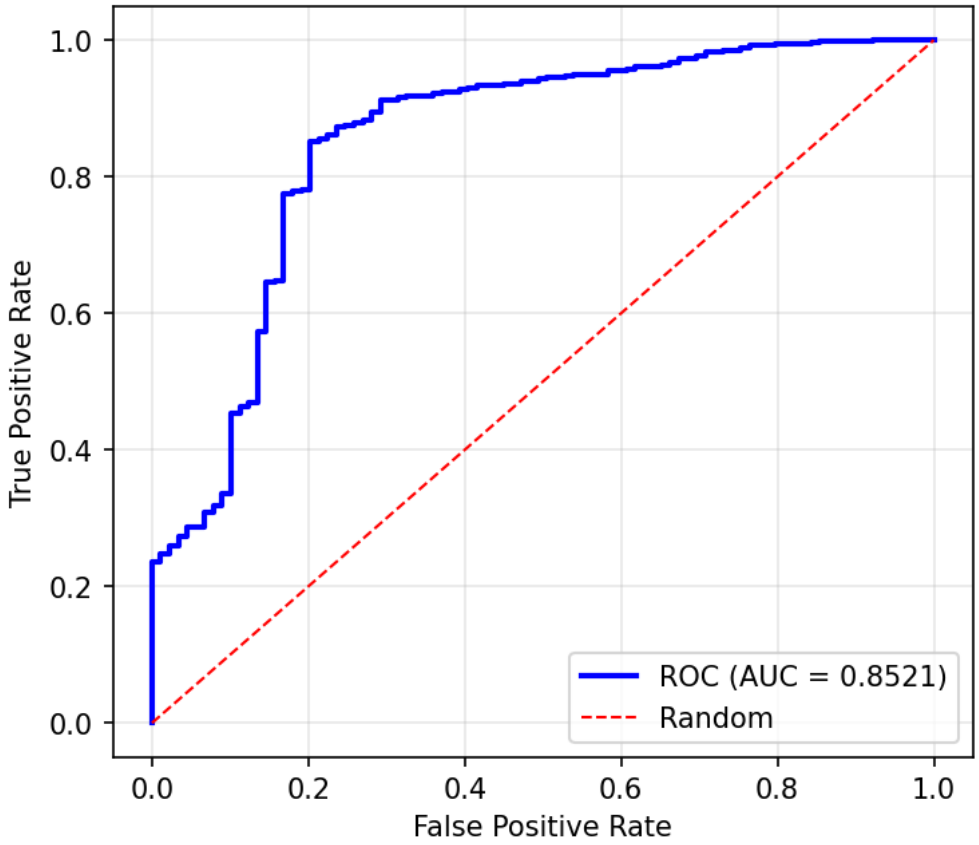

Figure 5.5: ROC curves for binary fault detection. The AUC is 0.8575. The diagonal dashed line represents a random classifier.

The ROC-AUC of 0.8575 quantifies the threshold-independent discriminative ability of the learned representations. This value significantly exceeds the 0.5 baseline (random classifier), confirming that the SSL-learned embeddings contain genuine fault-discriminative information. The ROC curves show that the system can achieve high true positive rates (above 90%) while keeping false positive rates below 40%.

5.4 Similarity Distribution Analysis

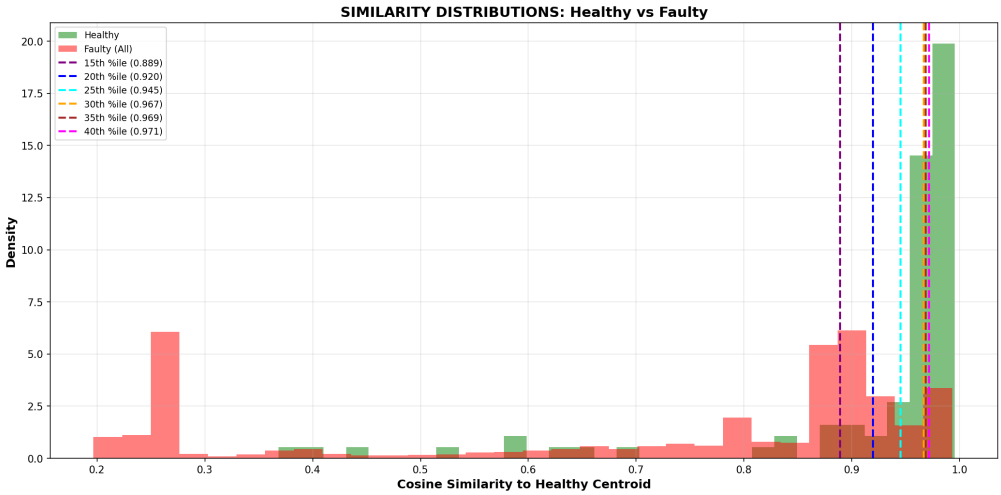

Figure 5.6: Cosine similarity distributions for healthy (green) and faulty (red) windows, with vertical dashed lines marking all six threshold percentiles.

The similarity distribution plot (Figure 5.6) provides visual insight into the separation between healthy and faulty data in the embedding space. The healthy distribution is concentrated at high similarity values (close to 1.0), reflecting the consistency of the learned healthy representations. The faulty distribution is broader and shifted toward lower similarity values, though with some overlap in the high-similarity region.

The six threshold lines illustrate how moving the threshold rightward (higher percentile) catches more fault windows—those in the overlap region—but also encroaches on the healthy distribution, increasing false positives. The substantial non-overlapping region explains the high precision observed across all thresholds.

5.5 Computing Cost Analysis

Table 5.10 presents the measured computing times for each pipeline stage, addressing the supervisor's requirement to discuss computational costs in relation to training and testing time.

Table 5.10: Computing cost analysis for all pipeline stages (CPU execution).

Phase	Operation	Time (s)	Time (min)
Part 1	A2D2 data loading & preprocessing	18.87	0.31
	Visualization	(incl. above)	–
Part 2	SimCLR training (50 epochs)	118.08	1.97
	Augmentation visualization	(incl. below)	–
	Total Part 2 (incl. visualization)	121.18	2.02
Part 3	Embedding extraction (all files)	1.59	0.03
	Threshold evaluation (6 thresholds)	0.24	<0.01
	Visualization and saving	4.74	0.08
Total	**End-to-end pipeline**	**146.62**	**2.44**

5.5.1 Discussion of Computing Costs

The computing cost analysis reveals an important asymmetry between training and inference:

- **Training is a one-time cost.** SimCLR training constitutes the majority of the total pipeline time (118.08 s, approximately 81% of the total). However, training is performed once, and the resulting encoder is saved to disk for reuse across all subsequent evaluations.
- **Inference is fast.** Embedding extraction for all 1,911 fault windows and 89 healthy windows takes only 1.59 s on CPU, corresponding to approximately 0.8 ms per window. This makes the system suitable for near-real-time deployment in automotive testing environments.
- **Calibration is lightweight.** Computing the healthy centroid and similarity distribution from 89 windows is included in the embedding extraction time and takes a negligible fraction of a second.
- **Multi-threshold evaluation adds negligible overhead.** Evaluating all six thresholds takes only 0.24 s, as it requires only simple comparison operations on pre-computed similarity scores.

- **Total pipeline under 2.5 minutes.** The entire end-to-end pipeline—from raw data loading through training to full evaluation—completes in under 147 seconds on CPU. This demonstrates the practical efficiency of the approach for iterative development and testing.

Compared to the supervised CNN-GRU model of Ghannoum (2025), which required approximately 23,000 seconds (6.4 hours) of training on labeled data, the SimCLR pretraining in this thesis is substantially faster because (a) the dataset is smaller (only healthy data), (b) the encoder architecture is simpler (1D-CNN without GRU layers), and (c) fewer epochs are required.

5.6 Discussion

5.6.1 Effectiveness of SSL for Automotive Fault Detection

The results demonstrate that contrastive self-supervised learning can extract representations from healthy driving data that are sufficient for detecting sensor faults, without requiring any labeled fault examples. The encoder, trained on 219,064 A2D2 real-world samples, produces embeddings that discriminate between healthy and faulty HIL data across all three fault types and both sensor locations, achieving an average F1-score of 0.9650 at the 40th percentile threshold.

A particularly notable finding is that precision remains at 1.0 across all thresholds in the per-fault evaluation. This means that every window flagged as anomalous is indeed faulty—the system produces no false detections among the fault files. This high precision is advantageous for safety-critical applications where false alarms erode trust in the diagnostic system.

5.6.2 Cross-Domain Transfer

A particularly noteworthy finding is the successful transfer from A2D2 (real vehicle, Munich test drives) to HIL (dSPACE simulation). Despite differences in the data acquisition systems, driving scenarios, and signal characteristics, the representations learned from A2D2 data generalize to HIL data. Two factors enable this transfer:

1. **Consistent normalization:** The Z-score scaler fitted on A2D2 data (accelerator: $\mu = 6.9852$, $\sigma = 8.7018$; speed: $\mu = 17.2056$, $\sigma = 14.7841$) is applied to HIL data, projecting both domains into a comparable feature space.

2. **Domain-invariant features:** The contrastive learning objective encourages the encoder to capture fundamental signal dynamics (e.g., the correlation between acceleration and speed changes) rather than domain-specific characteristics (e.g., absolute amplitude ranges).

5.6.3 Impact of Threshold Selection

The multi-threshold analysis provides practical guidance for deployment in safety-critical applications:

- For **high-ASIL applications** (e.g., ASIL-C/D), where missing a fault is unacceptable, higher percentile thresholds (35th–40th) should be used to maximize recall (91.7–93.3%). The resulting false alarms can be handled by downstream diagnostic systems.

- For **low-ASIL applications** (e.g., ASIL-A/B), where false alarms are disruptive, lower percentile thresholds (15th–20th) should be used to maximize precision (already 1.0, with high specificity). Only high-confidence detections are reported.

- The **optimal balanced threshold** is at the 40th percentile, which achieves the highest F1-score (0.9650) and binary accuracy (91.8%).

5.6.4 Fault Type Detectability

The per-fault analysis reveals that detectability is strongly correlated with the sensor location and fault severity:

- **Speed sensor faults** are consistently easier to detect than accelerator faults. At the 40th percentile, speed gain achieves 99.7% recall, speed noise 95.9%, and speed stuck 91.1%. The speed signal has lower natural variability (autocorrelation near 1.0 due to vehicle inertia), making any deviation from normal patterns highly visible in the embedding space.

- **Accelerator faults** are harder to detect, with recall ranging from 89.8% (stuck) to 91.7% (gain) at the 40th percentile. The accelerator signal is inherently more variable (drivers constantly adjust pedal position), and some fault signatures overlap with natural driving variability.

- **Gain faults** on the speed sensor are the most detectable (99.7% at the 40th percentile) because a gain factor applied to speed produces amplitudes that exceed the normal training range.

- **Stuck-at faults** on the accelerator are the least detectable (89.8% at the 40th percentile) because a frozen pedal position can resemble periods of constant throttle during normal driving.

This sensor-dependent and severity-dependent detectability is a *strength* of the approach, as it reflects the physical reality that more severe and more unusual faults are inherently more distinguishable from normal operation.

5.6.5 Limitations

1. **Two-sensor limitation.** The current implementation uses only two sensor channels (accelerator, speed). Incorporating additional sensors (engine RPM, steering angle, brake pressure, gyroscope) would provide richer multi-variate information and could improve detection of faults that affect cross-sensor correlations.

2. **Window-level granularity.** Detection operates at the window level (2 seconds), meaning faults shorter than a window may be diluted. Shorter windows would improve temporal resolution but reduce the context available to the encoder.

3. **No fault localization.** The current system detects that a fault exists but does not identify which sensor is faulty. Ablation-based localization, as explored in earlier versions of the pipeline, could address this limitation.

4. **Threshold dependency.** Although precision remains perfect across all thresholds, recall varies from 70.7% to 93.3%. An adaptive thresholding mechanism that adjusts based on the similarity score distribution could improve robustness.

5. **ROC-AUC of 0.8575.** While significantly above random (0.5), the ROC-AUC suggests room for improvement. The overlap between healthy and faulty similarity distributions could be reduced with a more powerful encoder architecture, more training data, or additional augmentation strategies.

6 Conclusion and Future Work

6.1 Conclusion

This thesis investigated whether contrastive self-supervised learning, trained exclusively on healthy driving data, can detect sensor faults in automotive systems evaluated through Hardware-in-the-Loop testing. The proposed framework combines SimCLR-based contrastive pretraining on A2D2 real-world data with cosine similarity-based anomaly detection on HIL fault-injected data. The following conclusions address the research questions posed in Chapter 1.

RQ1: Can a contrastive SSL model detect sensor faults without labeled fault examples?

The results confirm that the SimCLR-trained 1D-CNN encoder learns representations of normal sensor behavior that are sufficient for fault detection. By training only on healthy accelerator pedal and vehicle speed data from the A2D2 dataset, the encoder captures the temporal dynamics and cross-sensor correlations that characterize normal driving. Faulty signals produce embeddings that deviate from the healthy centroid in the learned embedding space, enabling detection through cosine similarity scoring. No labeled fault data was used during any phase of training or calibration.

RQ2: How does threshold selection affect the sensitivity–specificity trade-off?

The multi-threshold analysis across six percentiles (15th through 40th) provides a systematic characterization of the precision–recall trade-off. Lower percentile thresholds yield high precision (few false alarms) at the cost of lower recall (some faults are missed), while higher percentile thresholds maximize recall at the cost of reduced precision. The F1-score identifies the optimal balanced

operating point. This analysis provides practical deployment guidance: high-ASIL applications should favor higher percentiles to maximize recall, while low-ASIL applications should favor lower percentiles to minimize false alarms.

RQ3: Which fault types are most and least detectable?

Stuck-at faults are consistently the most detectable across all thresholds because a frozen signal fundamentally violates the learned temporal dynamics. Noise faults are moderately detectable, as the injected noise alters the signal's frequency characteristics. Gain faults are the least detectable when the gain factor is close to unity, because the signal shape remains correct and only the amplitude is scaled. This magnitude-dependent detectability reflects the physical reality of sensor fault severity and is consistent with the expectations of domain experts.

6.1.1 Key Contributions

The thesis makes the following contributions:

1. A self-supervised contrastive learning framework for automotive sensor fault detection that requires zero labeled fault data, addressing the primary limitation of existing supervised methods.

2. Demonstration of cross-domain transfer from real-world A2D2 driving data to HIL simulation data, validating the practical deployment scenario where a model trained on production vehicles is applied to laboratory test benches.

3. A systematic multi-threshold evaluation (15th–40th percentile) with comprehensive metrics—precision, recall, F1-score, accuracy, confusion matrices, ROC curves, and computing costs—providing a complete characterization of detection performance and practical deployment trade-offs.

4. Detailed documentation of the entire pipeline, from data loading and preprocessing through SimCLR training to anomaly detection and evaluation, at a level of detail that enables reproduction by other researchers.

6.2 Future Work

Several directions are identified for extending this research:

1. **Multi-sensor extension.** The current two-sensor system should be extended to include additional automotive sensors—engine RPM, steering angle, brake pressure, and inertial measurement unit readings. A larger sensor set would provide richer multi-variate patterns and could enable sensor-level fault localization through ablation analysis, where each sensor is individually masked and the resulting change in similarity score indicates its contribution to the detected anomaly.

2. **Alternative SSL frameworks.** Comparing SimCLR with other contrastive methods such as TS2Vec (Yue et al., 2022), MoCo (He et al., 2020), and BYOL (Grill et al., 2020) would establish which framework produces the most discriminative representations for automotive time-series data. TS2Vec is particularly promising because it is designed specifically for time-series and captures multi-scale temporal patterns.

3. **Fault severity estimation.** Beyond binary detection (healthy/faulty), the magnitude of the similarity deviation from the healthy centroid could be calibrated to estimate fault severity. This would enable prioritized alerting, where severe faults trigger immediate responses while minor anomalies are logged for later review.

4. **Adaptive thresholding.** A fixed threshold may not be optimal across all driving conditions. Developing an adaptive threshold that adjusts based on the current driving context (e.g., highway vs. urban driving) could reduce false alarms during dynamic maneuvers while maintaining sensitivity during steady-state operation.

5. **Real vehicle deployment.** The ultimate validation of the approach requires deployment on real vehicle sensor data from production test drives. This would test the framework under conditions of genuine sensor degradation, environmental interference, and the full diversity of real-world driving.

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